The joint resource question for causal recovery from a homogeneous tracial partial-SWAP stream

Exact acquisition accounting, physical processing obstructions, and the remaining optimal-memory problem

Astra research continuation and reproducibility record

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Abstract

The full optimal memory/processing law is not resolved in this continuation. The preserved all-prefix crossing converse, positive general-instrument witness, and exact pre-access loss law remain valid. A changing-reference Hilbert-space representation now gives a nonnegative, exact acquisition-loss ledger for every permitted causal CPTP receiver, including nontracial initialized workspace and unlimited classical feedback. Its final value determines an attained Petz-twirled recovery error and an all-decoder lower bound within a factor two. Consequences include a quantum retention-duration bound, a joint finite-count strengthening of the crossing converse, and a dimension-independent error floor for uniformly weak carrier-dependent readers that discard spent carriers. A Yangian Dirichlet form identifies fixed-prefix fourth-order leakage, with a rank-sensitive spin-sector constraint. A positive three-carrier instrument has an exact physical decoder and a paid packed-memory rate approaching 0.730 qubits per carrier, but has a nonzero accuracy floor. A new smooth energy-window instrument instead gives arbitrary-accuracy physical compression with exact angle/count bounds, actual support embeddings and a reverse decoder. An elementary version proves a strict weak-angle memory-coefficient improvement over the full-storage inverse reader, after paying extra acquisition and circuit service; its generic block processing is exponential and its retained width still has quartic angle scaling. Initialized coherent frames expose a distinct slow dynamics and an all-width phase-reference converse, with an explicit finite-angle lift. An exact polynomial certificate identifies why scalar normalization of the existing crossing argument cannot close its logarithmic gap. Exact finite-policy optimization is attainable in principle but has not been evaluated into the requested scaling. All limits and remaining construction gaps are stated explicitly.

1 One endpoint, unchanged physical premises

An unknown qubit S, possibly entangled with an inaccessible reference R, meets independent carriers E_j in $\tau = I/2$. Every source encounter is

$$U_j = cI + is\,\mathrm{SWAP}_{S,E_j}, \qquad c = \cos\theta,\ s = \sin\theta, \quad 0 < \theta < \pi/2.$$

The receiver accesses outputs once, chronologically; the final source is discarded. Arbitrary initialized workspace, CPTP instruments, classical histories and feedback are allowed. Every coherent buffer counts. No bath purifier, preparation key, return link or changed emitter is supplied. Error is half diamond distance from the identity channel.

The endpoint is the attainable joint relation among θ, ϵ, N, peak coherent storage and processing, with preparation, service, protection and finite control accounted for. A restricted-reader obstruction, a fixed-angle asymptotic, or an implicit optimum definition is not this full endpoint.

The earlier Astra acquisition calculation and completed peer spectral-profile converse are inputs [A1,A2]. Their preserved extension [A3] is included as an unchanged proof-note appendix and a separate original file in the archive. It contains the crossing proof, refined positive-instrument witness, exact carrier-loss law and physical reverse decoder, with their prior attribution.

Put $b = s^4$, $r = 1 - b$, $\ell = -\log r$. Storing all N outputs and applying inverse collisions to an independent mixed dummy in reverse chronology gives

$$\mathcal{D}_N \mathcal{C}_N = (1 - r^N)\,\mathrm{id} + r^N \mathcal{R}_{I/2}, \qquad \epsilon_P = \tfrac{3}{4} r^N, \tag{1}$$

$$\epsilon_{\mathrm{opt}} \geq \tfrac{3}{4}(1 - \sqrt{1 - r^N}). \tag{2}$$

Thus $N = \Theta(\theta^{-4} \log(1/\epsilon))$ jointly at weak angle and high accuracy. Source forgetting has visibility c^{2N} and a quadratic clock. At each fixed nonzero angle the preserved general bound is

$$q_{\mathrm{peak}} = \Omega_\theta\left(\frac{\log(1/\epsilon)}{\log\log(1/\epsilon)}\right) = \Omega_\theta(N/\log N)$$

at the benchmark, versus the known $O(N)$ upper bound. The logarithmic gap and the weak-angle fixed-error optimum remain open here.

2 An exact positive ledger for every causal receiver

Let $\mathcal{N}_j : L \to SM_j$ be the channel after update j, where L is the initial logical qubit. Include all classical histories in M_j as a direct sum, and all initialized quantum workspace. The exact invariant is

$$\mathcal{N}_j(\tau) = \tau_S \otimes \sigma_j. \tag{3}$$

For tracial logical input, each fresh SE_j pair is tracial and independent of the receiver before its collision. The source unitary leaves this average unchanged; the receiver then acts only on $M_{j-1}E_j$. This proves the product form inductively, including conditional classical blocks. It asserts no such independence for nontracial input or for the Bell state with its reference.

For the three Pauli directions define balanced operators

$$X_{\mu,j} = (\tau_S \otimes \sigma_j)^{-1/4} \mathcal{N}_j(\sigma_\mu/2)(\tau_S \otimes \sigma_j)^{-1/4}.$$

All inverses are on support; positivity puts the logical images on that support. Use ordinary Hilbert–Schmidt norms, summing classical blocks. Let $P(X) = I_S \otimes \mathrm{Tr}_S(X)/2$, $Q = I - P$, and

$$a_j = \frac{1}{3}\sum_\mu \|P X_{\mu,j}\|_2^2\,, \qquad q_j = \frac{1}{3}\sum_\mu \|Q X_{\mu,j}\|_2^2\,.$$

Here q_j is an information norm, not a qubit count. Initially $a_0 = 0, q_0 = 1$ for any independent receiver initialization.

Theorem 1 (All-receiver acquisition ledger)**.** *Every permitted causal receiver has nonnegative losses ℓ_j^P, ℓ_j^Q with the exact relations*

$$a_j = a_{j-1} + b q_{j-1} - \ell_j^P, \qquad q_j = r q_{j-1} - \ell_j^Q, \tag{4}$$

$$a_j + q_j = 1 - \sum_{k \leq j} (\ell_k^P + \ell_k^Q),$$

$$a_N = 1 - r^N - L_N, \qquad L_N = \sum_{j=1}^{N} [\ell_j^P + (1 - r^{N-j})\ell_j^Q]. \tag{5}$$

The actual memory channel $\mathcal{E}_N = \mathrm{Tr}_S \mathcal{N}_N$ has Petz entanglement fidelity $(1+3a_N)/4$. Every final decoder obeys

$$\epsilon_{\mathrm{opt}} \geq \tfrac{3}{4}(1-\sqrt{a_N}). \tag{6}$$

A finite global Clifford twirl of the receiver and its physical Petz decoder attain

$$\epsilon_{P,\mathrm{twirl}} = \tfrac{3}{4}(1-a_N) = \tfrac{3}{4}(r^N + L_N). \tag{7}$$

Petz is CPTP here; cheap synthesis and fast service are not assumed.

Proof. For a CPTP map Λ taking average ω to σ, the relative map

$$V(X) = \sigma^{-1/4}\Lambda(\omega^{1/4} X \omega^{1/4})\sigma^{-1/4}$$

is a Hilbert–Schmidt contraction. This is the monotone quantum χ^2 metric with $f(t) = \sqrt{t}$ [1], or follows by expanding sandwiched Rényi-2 data processing [2,3] at $\omega + tA$. Trace preservation cancels the constant and linear terms. A trace normalization, when needed, cancels on both sides. Hermitian and anti-Hermitian parts give the general contraction. Continuous classical outcomes are handled by direct integrals in place of direct sums. On SME it acts as $\mathrm{id}_S \otimes V$, preserving and contracting P, Q separately, including all classical direct sums.

Appending τ_E appends $I_E/\sqrt{2}$ in balanced coordinates. The source unitary commutes with the average and therefore acts unitarily. The accepted collision orthogonality [A1] gives $a_j^- = a_{j-1} + s^4 q_{j-1}$ and $q_j^- = r q_{j-1}$. Newly exported source-identity terms have a traceless fresh-carrier factor and are orthogonal to old exports. All old receiver coefficients remain in this identity. Set $\ell_j^P = a_j^- - a_j$, $\ell_j^Q = q_j^- - q_j$. Contractivity proves nonnegativity; iteration proves (5).

Pauli expansion of the normalized Choi state expresses Petz Bell fidelity as one quarter of four balanced squared norms. The identity term is one; tracing S selects precisely P. Thus $F_P = (1+3a_N)/4$.

For a final decoder D, let

$$y_\mu = \sigma^{-1/4}\mathcal{E}_N(\sigma_\mu/2)\sigma^{-1/4}, \quad z_\mu = \sigma^{1/4} D^*(\sigma_\mu)\sigma^{1/4}.$$

Then $F_e(D\mathcal{E}_N) = 1/4 + \sum_\mu \langle z_\mu, y_\mu\rangle/4$. For Hermitian B, $\mathrm{Tr}(\sqrt{\sigma}B\sqrt{\sigma}B) \leq \mathrm{Tr}(\sigma B^2)$ by the nonnegative squared commutator with $\sqrt{\sigma}$. Kadison's inequality gives $D^*(\sigma_\mu)^2 \leq I$, so $\|z_\mu\|_2 \leq 1$. Cauchy–Schwarz and $\sum \|y_\mu\|_2^2 = 3a_N$ prove (6). This direct check retains the prior attribution to general sharp recovery comparisons [4].

Global source covariance permits a common random Clifford on every incoming carrier, a classical record and the inverse output rotation. This twirls the logical channel without changing its entanglement fidelity, making it depolarizing with visibility a_N. Equation (7) follows; its ratio to the lower bound is $1 + \sqrt{a_N} \leq 2$. □

The resulting joint budgets, for $0 \leq \epsilon \leq 3/4$, are

$$L_N \leq 1 - r^N - (1-4\epsilon/3)^2 \qquad \text{necessary,} \tag{8}$$

$$r^N + L_N \leq 4\epsilon/3 \qquad \text{sufficient with Petz and twirl.} \tag{9}$$

The central unknown is the attainable minimum of this weighted loss under memory and processing constraints. The identity alone does not evaluate it.

3 The preserved converse acquires an exact time interpretation

At prefix n, set $D = 2^n$ and $H_n = I - T_n T_n^*$ for the tracial bath channel. A refined instrument has $A_x = (K_x^\dagger K_x)^{1/2} \geq 0$, $\operatorname{rank} A_x \leq m$ and $\sum A_x^2 = I$. The preserved witness is

$$Z_n = D^{-1} \sum_x |A_x\rangle\!\rangle\langle\!\langle A_x|, \quad \operatorname{Tr} Z_n = 1, \quad Z_n \leq (m/D) I, \quad \operatorname{Tr}(H_n Z_n) \leq 2\epsilon - \tfrac{4}{3}\epsilon^2.$$

It also has tracial marginals and Schmidt number at most m [A2,A3]. For a fully Kraus-refined physical receiver,

$$\operatorname{Tr}(H_n Z_n) = \tfrac{3}{4} \sum_{j \leq n} (\ell_j^P + \ell_j^Q).$$

For a coarse receiver, extra Kraus labels are revealed only in a converse relaxation, and the correct relation is $\leq$ instead of equality. This distinction prevents attributing an unrealized refinement's losses to the actual physical record.

With $u = \cot\theta$, the preserved crossing lower bound is

$$B_n = (u^2 + 4n^2)/u, \quad K_n = \frac{B_n^n[(u+2n)^n + 1]}{n!}, \quad \gamma_n \geq g_n = \frac{6}{n(n+1)(2n+1)K_n^2} > 0.$$

Every prefix loss has final weight at least $1 - r^{N-n}$. Thus for $n < N$ every successful receiver satisfies the finite-parameter condition

$$\frac{m}{2^n} \geq 1 - \frac{3\{1 - r^N - (1 - 4\epsilon/3)^2\}}{4(1 - r^{N-n}) g_n}. \tag{10}$$

A negative numerator violates even the all-output acquisition condition; a negative right side is vacuous. The angle, accuracy and count quantifiers are retained, but this g_n does not close the joint scaling gap.

4 A general quantum-retention duration bound

Let π_j be the tracial-input probability that the actual branch at cut j has quantum memory dimension at least two. Put

$$A_\epsilon = (1 - 4\epsilon/3)^2, \qquad C_\epsilon = \tfrac{3}{2}(1 - A_\epsilon) = 4\epsilon - \tfrac{8}{3}\epsilon^2.$$

Every successful receiver obeys

$$\pi_j \geq [1 - C_\epsilon r^{-(N-j)}]_+. \tag{11}$$

For proof, a scalar quantum branch has effect E_x, probability $p_x = \operatorname{Tr} E_x/2$, and Petz Bell contribution $\operatorname{Tr} E_x^2/(4 \operatorname{Tr} E_x) \leq p_x/2$. An arbitrary quantum branch contributes at most p_x. Hence $(1 + 3a_j)/4 \leq (1 + \pi_j)/2$. Future acquisition is bounded by $a_j + (1 - r^{N-j}) q_j \leq 1 - r^{N-j}(1 - a_j)$; combine with $a_N \geq A_\epsilon$ and rearrange.

If every branch is quantum-free at a cut and $C_\epsilon < 1$, at least $\log(1/C_\epsilon)/\ell$ encounters must remain. For spacings Δ_j, expected quantum-memory time is at least

$$\sum_{j=1}^{N-1} \Delta_j [1 - C_\epsilon r^{-(N-j)}]_+.$$

The same bound holds at every time between arrivals: insert receiver-only operations as zero-acquisition steps, keeping the same remaining collision count, and integrate. Count any output

already produced but held until the declared handoff; an earlier completed task uses its actual earlier deadline. For constant Δ and a long enough horizon, its full positive part is

$$\Delta\left[\frac{\log(1/C_\epsilon)-1+C_\epsilon}{\ell}+O(1)\right],$$

with sum–integral error at most Δ. This earns the joint lower bound $\Omega(\Delta\theta^{-4}\log(1/\epsilon))$ on expected qubit-time. It does not give peak width or identical reset times in every history.

5 A general obstruction to uniformly weak carrier processing

The algebraic constructor found an environmental witness for coherent crossed flags. Its following extension allows arbitrarily large initialized memory, all flag coherences, feedback and strong retained-register controls. The restriction is the actual carrier-dependent update before a spent carrier is discarded.

Dilate that update as $W_j : M_jE_j \to M'_jE_j$, with all retained flags and workspace in M'_j. Suppose a receiver-only isometry V_j satisfies

$$\|W_j - V_j \otimes I_E\|_\infty \le \delta_j. \tag{12}$$

The condition is uniform over histories and refers to the whole dilation, not a marginal or postselected branch. Other receiver environments may be retained in a stronger converse receiver. Passive carrier storage and a swap into a buffer are not weak discarded-carrier updates in this sense.

Theorem 2 (Finite processing-strength bound)**.** *Examine n such steps after any k preceding collisions, during which the receiver may have acted arbitrarily. Let $v_\theta = 2\sin(\theta/2)$ and*

$$h_j = cs + 2\delta_j + 4\delta_j v_\theta, \qquad H = \sum_{j=1}^{n} h_j, \tag{13}$$

$$B = 2cs\,c^{2k}\frac{1-c^{2n}}{s^2} - 4\sqrt{3}s^2\sum_{j=1}^{n}\delta_j\sqrt{k+j-1} - 8v_\theta\sum_{j=1}^{n}\delta_j.$$

Every subsequent continuation has recovery error

$$\epsilon \ge \frac{[B]_+^4}{256(n+H^2)^2}. \tag{14}$$

If $k=0$, $\delta_j \le \kappa\theta$ for fixed finite κ, and $n = \lfloor \alpha/\theta^2 \rfloor$ for fixed $\alpha > 0$, this implies

$$\liminf_{\theta\to 0}\epsilon \ge \frac{(1-e^{-\alpha})^4}{16\alpha^2\{1+\alpha(1+2\kappa)^2\}^2} > 0. \tag{15}$$

No bound on memory dimension is used.

Proof. Purify each fresh carrier using an inaccessible F_j in $\Phi_{E_jF_j}$. On the spent inaccessible pair choose $O_j = i[Z_{E_j}, \Phi_{E_jF_j}]$, with norm one. Distinct O_j commute. Put $Q = W^\dagger OW - O$, so $\|Q\| \le 2\delta$, and $B_M = \langle\Phi|Q|\Phi\rangle$, which acts only on receiver memory and has norm at most 2δ. Expanding the source gate gives exactly

$$\langle\Phi|U^\dagger OU|\Phi\rangle = csZ_S.$$

The squared terms vanish; the cross term gives this source Pauli. The full effective pre-collision observable is therefore $csZ_S + B_M + R$, where $\|R\| \le 2\,\|U - I\|\,\|Q\| \le 4\delta v_\theta$. Its norm is at most h_j. No smoothness of the controls in θ is assumed.

Logical inputs $|0\rangle, |1\rangle$ have source polarization $\pm c^{2t}$ after t collisions. For any receiver observable C, balanced Cauchy–Schwarz gives

$$|\operatorname{Tr} C\{\mathcal{E}_t(|0\rangle\langle 0|) - \mathcal{E}_t(|1\rangle\langle 1|)\}| \le 2\,\|C\|\,\sqrt{3a_t}, \qquad a_t \le ts^4.$$

The source term, this receiver signal bound and the remainder make the difference in means of $Z = \sum_j O_j$ at least B.

Condition on earlier converse measurements of the environmental observables. Each next Bell pair remains fresh; the conditional mean is bounded by h_j. The martingale differences have second moment at most n, while the predictable drift has magnitude at most H. The sum of both input second moments is at most $4n + 4H^2$. Weighted Cauchy–Schwarz for two probability distributions therefore gives

$$D_{\text{env}} \ge [B]_+^2/[8(n + H^2)].$$

Bell fidelity at least $1 - \epsilon$ makes the complementary Choi state within trace distance $\sqrt{\epsilon}$ of a product with the logical reference, by purification and Uhlmann's theorem. Measuring that reference in the Z basis and using the triangle inequality gives $D_{\text{env}} \le 2\sqrt{\epsilon}$. This is standard information–disturbance machinery [5]. It proves (14). Finally $\theta B \to 2(1 - e^{-\alpha})$ and $\limsup \theta H \le \alpha(1 + 2\kappa)$, proving (15). □

If the arbitrary preceding prefix has $k\theta^2 \to \beta < \infty$, the limiting numerator gains $e^{-4\beta}$. If β or κ diverges, use the exact finite bound instead. For a predetermined fraction $f > 0$ of weak steps within the first α/θ^2 encounters, the same proof gives

$$\liminf \epsilon \ge \frac{f^2\alpha^2 e^{-4\alpha}}{16\{1 + f\alpha(1 + 2\kappa)^2\}^2},$$

even if all other steps use arbitrary strong operations. The proof does not license choosing the tested subset using a hidden input-dependent history.

In an interaction picture removing receiver-only controls, Duhamel's bound gives, with $\hbar = 1$,

$$\delta_j \le \int \|H_{\text{int}}(t)\|\, dt.$$

Thus uniform carrier-coupling action $O(\theta)$ has the stated accuracy floor. This is a processing condition; it is not a width converse or a claim that strong processing suffices. It permits a nonzero amount of acquired information in a fixed-memory weak-angle limit.

5.1 Coherent crossed flags still lose the payload

The earlier candidate uses

$$W = |0\rangle K + |1\rangle L, \quad K = \frac{cI - 2is\Phi}{\sqrt{1 + 3s^2}}, \quad L = \frac{2s(I - \Phi)}{\sqrt{1 + 3s^2}},$$

and discards the spent carrier. With Bell-initialized payload and a retained purification, the full coherent retained channel is a complement of a depolarizing spent-carrier channel of visibility $(1 + s^2)/(1 + 3s^2)$. Its exact Petz visibility is

$$\lambda = \frac{s\sqrt{2 + 3s^2} + s^2}{1 + 3s^2} = \sqrt{2}\,\theta + O(\theta^2).$$

Indeed the complementary Pauli Kraus weights are $p_0 = (2+3s^2)/[2(1+3s^2)]$ and $p_a = s^2/[2(1+3s^2)]$; balancing their identity/Pauli and Pauli/Pauli coherences gives $\lambda = 2\sqrt{p_0 p_a} + 2p_a$. The first update has $\ell_1^Q = 2(1-\lambda)c^2s^2$. Later coherent flag processing cannot restore that loss.

For this particular W, the effective environmental observable can be calculated more sharply. With $D = 1+3s^2$, it is

$$\frac{cs(1+s^2)}{D} Z_S - \frac{c^3 s}{D} Z_M + \frac{c^2 s^2}{2D}(X_S Y_M + Y_S X_M), \qquad \|A\| = \frac{2cs}{D}.$$

The preceding martingale proof over α/θ^2 updates gives

$$\liminf \epsilon \geq \frac{(1-e^{-\alpha})^4}{16\alpha^2(1+4\alpha)^2}.$$

At $\alpha = 1/2$ this is about 6.66×10^{-4}. The full exact finite-angle and priming formulas are retained in the algebraic audit. An occupation tail for coherent flags cannot be combined with high-fidelity recovery for this spent-carrier-discarding architecture.

6 Independent local preprocessing has a separate exact law

Pre-access erasure remains distinct from coherent processing before discard. For independent local CPTP preprocessors $\mathcal{L}_j : E_j \to B_j$, with retained local classical records but no shared coherent register or temporal feedback, suppose each local Petz composition is depolarizing with visibility λ_j. Then exactly

$$a_j = a_{j-1} + \lambda_j s^4 q_{j-1}, \qquad q_j = (c^4 + 2\lambda_j c^2 s^2) q_{j-1}.$$

The physical composite Petz decoder has visibility a_N and the all-decoder bound (6) applies. To prove the recurrence, resolve each source Pauli into the three orthogonal groups of weights $s^4, c^4, 2c^2s^2$, and apply the local relative-norm map. This retains arbitrary coefficients on old outputs; it assumes no independence of those signal coefficients.

For homogeneous λ, writing $f = c^4 + 2\lambda c^2 s^2$ gives

$$a_N = \lambda s^4 \frac{1-f^N}{1-f}, \qquad a_\infty = \frac{\lambda s^2}{s^2 + 2(1-\lambda)c^2}.$$

Necessary accuracy ϵ requires

$$1 - \lambda \leq \frac{(1-A_\epsilon)s^2}{s^2 + 2A_\epsilon c^2}.$$

Petz sufficiency replaces A_ϵ by $1 - 4\epsilon/3$ and allows a finite-count margin. Local Petz loss must consequently be $O(\epsilon\theta^2)$ jointly at weak angle and high accuracy. Flagged retention probability p has $\lambda = p$ and reproduces the earlier exact erasure law; unflagged depolarization of visibility v has $\lambda = v^2$. Shared coherent processing is governed by the general ledger, not these product formulas.

7 The algebraic opening: collective spin and ordered current

Both the channel and algebraic constructions independently found

$$J^a = \sum_{j=1}^{n} \sigma_j^a, \qquad Y^a = \sum_{j<k} (\boldsymbol{\sigma}_j \times \boldsymbol{\sigma}_k)^a, \qquad \mathcal{A}_n = \{X : [J^a, X] = 0 \ \forall a\}.$$

The algebra $\mathcal{A}_n$ contains all SU(2) multiplicity coherences. At each *fixed* n the leakage form satisfies

$$\theta^{-4}(I - T_n^* T_n)|_{\mathcal{A}_n} \longrightarrow \mathcal{L}_n, \quad \langle X, \mathcal{L}_n X\rangle = \frac{1}{16}\sum_a \|[Y^a, X]\|_2^2 . \tag{16}$$

The collision product has logarithm $i\theta\sum_j P_{S,j} + (i\theta^2/4)\sum_a \sigma_S^a Y^a + O_n(\theta^3)$. The first term commutes with $I_S \otimes X$; source-traceless leakage starts with $(i\theta^2/4)\sum_a \sigma_S^a [Y^a, X]$, proving the form.

Up to fixed factors and a common evaluation shift, J, Y are the level-zero and level-one Yangian generators [6]. Molev's irreducibility criterion for identical fundamental evaluation modules [7] makes their simultaneous commutant scalar. Thus

$$\lambda_n = \min_{\substack{X\in\mathcal{A}_n, \operatorname{Tr} X=0 \\ \|X\|_2=1}} \frac{1}{16}\sum_a \|[Y^a, X]\|_2^2 > 0, \qquad \gamma_n(\theta) = \theta^4[\lambda_n + O_n(|\theta|)]$$

for fixed $n \geq 2$. Nonscalar operator-spin l sectors begin at $\theta^2 l(l+1)$ and cannot mix with the scalar sector. Numerically $\lambda_2 = 2$, $\lambda_3 = (7 - \sqrt{33})/2$ and $\lambda_{10} \approx 0.00815285$. The small-system trend is not a uniform bound; no extrapolation to $n \sim \theta^{-4}$ is licensed.

7.1 Positive instruments impose a stronger rank constraint

Write the Schur decomposition as $\bigoplus_j \mathbb{C}^{r_j} \otimes V_j$, $d_j = 2j+1$ and $w_j = d_j r_j / 2^n$. Let $\beta_n(\theta)$ be the smallest eigenvalue of H_n on nontrivial SU(2) operator sectors. Every positive rank-m refined instrument satisfies

$$\operatorname{Tr}(H_n Z) \geq \beta_n(\theta)\sum_j w_j(1 - m/d_j)_+ . \tag{17}$$

For proof, twirl a positive effect A onto the commutant. Positivity and rank at most m bound its block-j squared norm by

$$d_j^{-1}\left\|\operatorname{Tr}_{V_j}(P_j A P_j)\right\|_2^2 \leq \min(1, m/d_j)\,\|P_j A P_j\|_2^2 .$$

Sum the effects and use $\sum A_x^2 = I$; all remaining mass costs at least β_n. For fixed n, $\beta_n = 2\theta^2 + O_n(\theta^3)$. At $n = 2, m = 2$ the factor is $1/4$, giving energy at least $\theta^2/2 + O(\theta^3)$ even though the unconstrained signed gap is order θ^4. A soft signed mode alone does not supply a complete low-rank instrument. This stronger constraint still has no uniform causal evaluation sufficient for the endpoint.

Writing $D_n(m) = \sum_j w_j(1 - m/d_j)_+$, one also has

$$D_n(m) \geq \left[1 - m 2^{-n}\binom{n}{\lfloor n/2\rfloor}\right]_+ .$$

Separating the scalar identity from the other invariant operators strengthens the energy bound to

$$\operatorname{Tr} H_n Z \geq \gamma_n(1 - m/2^n) + (\beta_n - \gamma_n) D_n(m).$$

Combining the basic rank bound with the exact ledger yields, for any prefix $n < N$ of any successful causal receiver,

$$L_N \geq \tfrac{4}{3}(1 - r^{N-n})\beta_n(\theta) D_n(m).$$

These inequalities retain the exact angle and horizon factors; the missing uniform spectral estimate has not been replaced by a fixed-n expansion.

8 A physical block compressor and an exact recovery guarantee

The algebraic constructor found a positive instrument, not just a signed low-energy direction. On three consecutive carriers put

$$P = \tfrac{1}{4}I - \tfrac{1}{6}(\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 + \boldsymbol{\sigma}_2\cdot\boldsymbol{\sigma}_3) + \tfrac{1}{12}\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_3.$$

This is a rank-two orthogonal projector. Coupling the outer sites first, the sectors $(j_{13}, j) = (0, 1/2), (1, 1/2), (1, 3/2)$ give eigenvalues $0, 1, 0$. Measure only P versus $I-P$, retain the complete selected range, and store the binary outcome classically. The rank-six branch retains coherence between its spin-3/2 and spin-1/2 components. Measuring those components separately would define a different, more destructive receiver.

8.1 Exact acquisition and physical reversal

With $x = \sin^2\theta$, define

$$b_3 = \frac{x^2(x^4 + 30x^3 - 60x^2 - 12x + 54)}{18}, \tag{18}$$

$$h_3 = \frac{(1-x)^3(4x^3 + 24x^2 + 27x + 9)}{9}. \tag{19}$$

For K such independently processed triples, the exact ledger is

$$a_K = b_3\frac{1 - h_3^K}{1 - h_3}, \qquad q_K = h_3^K. \tag{20}$$

Here K counts blocks, so the original collision count is $N = 3K$.

For a direct proof let $U = U_3U_2U_1$ and

$$W_a = U(\sigma_a^S \otimes I)U^\dagger, \quad Y_a = (I\otimes P)W_a(I\otimes P) + (I\otimes(I-P))W_a(I\otimes(I-P)).$$

Exact Pauli multiplication gives

$$b_3 = \mathrm{Tr}[(\mathrm{Tr}_S\, Y_a/2)^2]/8, \quad b_3 + h_3 = \mathrm{Tr}[Y_a^2]/16,$$

and

$$1 - b_3 - h_3 = \frac{x^2(7x^4 - 6x^3 - 6x^2 + 4x + 6)}{18}.$$

The projectors and source are SU(2) covariant, so the cross Gram matrices are scalar multiples of δ_{ab}. The new source-identity term has zero trace on the new block and is orthogonal to old acquired information. These facts extend the scalar contractions to arbitrary old memory coefficients and references, proving (20). The exact symbolic contraction is included in the reproducibility archive.

For a branch isometry V_x from the selected range into its compressed space, the average state is $I_{\mathrm{range}}/8$. Its Petz inverse is the conditional embedding $Z_x \mapsto V_x^\dagger Z_x V_x$ into the original triple. Restore the last stored block, run its three inverse collisions on an independent mixed dummy from newest to oldest, discard the restored block, and continue backwards. Composition of the Petz maps for these consistent reference states proves that this is the physical Petz decoder of the actual processed stream. Covariance gives exactly

$$\epsilon_{3,P}(K,\theta) = \tfrac{3}{4}(1 - a_K). \tag{21}$$

No source access, preparation label or bath purification is used.

8.2 Deterministic and probabilistic capacity guarantees

Three processed triples have maximum branch dimension $6^3 = 216 < 2^8$. A conditional nine-qubit encoder therefore stores every branch in eight qubits, plus three classical bits. For N a multiple of nine, this gives $8N/9$ retained data qubits and a fixed raw-block workspace when ideal service keeps pace with arrivals. Fixed-size encoders and decoders use a finite, possibly large, constant number of gates per group, plus exactly N inverse source-angle collisions at the final sweep.

The operational constructor improved the capacity coefficient by packing the variable branch dimensions. Every full label effect is SU(2) invariant. The source adjoint sends it to a scalar on the input qubit, so its probability is independent of every logical input and inaccessible reference. On tracial input all carriers are product tracial. Hence the labels are exactly iid: the dimensions $2, 6$ occur with probabilities $1/4, 3/4$.

If X_K counts rank-two labels, the retained branch dimension and its mean logarithm are

$$D_K = 2^{X_K} 6^{K-X_K}, \qquad \mu = 1 + \tfrac{3}{4}\log_2 3 = 2.1887218755\ldots \quad \text{bits per triple.}$$

Hoeffding's inequality gives

$$\Pr\left[\log_2 D_K > K\mu + (\log_2 3)\sqrt{\tfrac{K}{2}\log(1/\delta)}\right] \le \delta.$$

An overflow branch may abort to a fixed output. Its input-independent probability makes the added half-diamond error at most δ, including reference inputs. This is a capacity guarantee; an average entropy alone would not justify it.

Pack groups of L triples, keeping their known classical radices. Rounding each group's logarithmic dimension adds at most $\lceil K/L \rceil$ qubits. One sufficient packed-data capacity is

$$W = \left\lceil K\mu + (\log_2 3)\sqrt{\tfrac{K}{2}\log(1/\delta)} + \lceil K/L \rceil \right\rceil. \tag{22}$$

Add $O(L)$ raw-buffer and arithmetic workspace. A coherent radix codec computes $z = ra + b$ and uncomputes a, b by reversible division, with $r \in \{2, 6\}$. A group costs $O(L^2)$ addressed arithmetic gates, so all packing and unpacking cost $O(NL)$. It preserves arbitrary superpositions within the selected ranges. Ceilings and the probability tail give the asymptotic packed data rate

$$\mu/3 = 0.7295739585\ldots \quad \text{qubits per original carrier}$$

as K, L grow with $L = o(K)$ and $\log(1/\delta) = o(K)$. Fixed large L retains an explicitly paid $O(1/L)$ rate overhead and linear addressed work.

There is a real service bill. At fixed carrier spacing Δ, a packed group with bounded-speed service time τ_{group} requires enough processors or queue space to accommodate arrivals during that service. Fixed-size nine-carrier encoders compiled to total error δ_{ctrl} use $O(N \operatorname{polylog}(N/\delta_{\text{ctrl}}))$ gates; their extra coherent processor/queue cells can be polylogarithmic, not constant, at a fixed clock. The radix codec has the stated L-dependent service. Routing, zero-state preparation, pointer measurement/reset, finite classical control and protection of every occupied cell are additional explicit obligations. The final dummy sweep is sequential. The ideal CPTP model does not specify a hardware noise law from which a universal physical protection overhead could be inferred.

8.3 The operation's genuine accuracy boundary

Put $a_\infty = b_3/(1 - h_3)$ and $\epsilon_\infty = 3(1 - a_\infty)/4$. Exactly,

$$\epsilon_{3,P} = \epsilon_\infty + \tfrac{3}{4} a_\infty h_3^K.$$

Thus any target $\epsilon > \epsilon_\infty + \delta + \delta_{\mathrm{ctrl}}$ is attained by rounding up the corresponding logarithmic bound on K, with (22) and the paid workspace and processing above. At weak angle,

$$b_3 = 3\theta^4 + O(\theta^6), \quad h_3 = 1 - \tfrac{10}{3}\theta^4 + O(\theta^6), \quad a_\infty \longrightarrow 9/10.$$

In fact $a_\infty \le 9/10$ at every angle. Their difference equals

$$\frac{x(31x^3 - 42x^2 + 3x + 24)}{10(4x^4 + 12x^3 - 33x^2 - 4x + 30)}.$$

The denominator is positive because $1 - h_3 > 0$; the cubic has positive Bernstein coefficients $24, 25, 12, 16$ on $[0, 1]$. This proves the bound. The physical Petz error floor is therefore at least 0.075. Every final decoder on this processed stream has the lower floor $\frac{3}{4}(1 - \sqrt{0.9}) \approx 0.0384875$.

The packing saving is real but its additional acquisition time can offset it. For illustration, at fixed weak-angle target $\epsilon = 0.2$ and vanishing packing/control margins, its leading data capacity is $1.1073\theta^{-4}$, versus $1.3218\theta^{-4}$ for storing all outputs and using the simple inverse decoder. This compares two attained readers, not either reader with the unknown optimum. The quartic exponent and the arbitrary-accuracy endpoint have not been improved by this operation.

Numerical balanced positive cuts on six and seven sites have smaller fixed-block fourth-order losses and Petz floors about 0.03326 and 0.01517. Their saved matrices are feasible numerical instruments, not certified optima or a uniform family. A recursive operation must prove what representation each retained block carries before reusing it.

9 A uniform energy-window compressor with arbitrarily small error

Standard rational-XXX train algebra [18] yields a second physical positive instrument. Its accuracy increases with block size, while its retained-memory rate tends to one and its generic service cost grows exponentially. These costs remain attached to the construction.

For $n \ge 2$ consecutive carriers set $H_n = \sum_{j=1}^{n-1} P_{j,j+1}$ and $U = U_n \cdots U_1$. The exact boundary relation is

$$[H_n, U] = s^2(UP_{S,1} - P_{S,n}U), \qquad \|[H_n, U]\|_{2,\tau}^2 = \tfrac{3}{2}s^4.$$

Indeed for $L_j = uI + iP_{S,j}$, $u = \cot\theta$, permutation algebra gives $[P_{j,j+1}, L_{j+1}L_j] = -i(L_j - L_{j+1})$. Multiplying by the surrounding factors telescopes to the endpoints; $L_j^{-1} = (uI - iP_{S,j})/(u^2 + 1)$ gives the displayed relation. The endpoint terms have normalized trace overlap $1/4$, proving the norm.

Put $W_a = U(\sigma_a^S \otimes I)U^\dagger$. Positive complete effects A_x followed by support encoding have exact source-granted loss

$$d = 1 - \beta - h = \tfrac{1}{6}\sum_{a,x} \|[I_S \otimes A_x, W_a]\|_{2,\tau}^2 .$$

The charge quadratic form is exactly

$$D_U(H_n) := \tfrac{1}{6}\sum_a \|[I_S \otimes H_n, W_a]\|_{2,\tau}^2 = s^4(1 + r^{n-1}).$$

Pauli completeness rewrites it as $\frac{4}{3}\left\|Q_S[U^\dagger H_n U]\right\|_{2,\tau}^2$. The endpoint source-traceless squared norms are $3/4$ and $(3/4)r^{n-1}$, with zero cross term, giving the equality.

For a real nonnegative unit vector $F(E) = (f_x(E))$ with Euclidean Lipschitz constant L_F, set $A_x = f_x(H_n)$. Matrix elements in the energy basis show

$$\sum_x \|[f_x(H_n), W]\|_2^2 \le L_F^2 \, \|[H_n, W]\|_2^2 , \quad d \le \min\{1, L_F^2 s^4 (1 + r^{n-1})\}.$$

This is a Hilbert–Schmidt inequality. Coarse overlapping cosine windows of width $2w$ and grid spacing w have squares summing to one and $L_F = \pi/(2w)$. They retain all coherences inside each support.

9.1 Rank bounds and an elementary two-outcome version

At tracial input H_n has mean $(n-1)/2$ and variance $3(n-1)/4$. The product-state Berry–Esseen theorem [15, Theorem 1] gives Kolmogorov error $K_{\mathrm{BE}}/\sqrt{n}$ for these bounded nearest-neighbor terms. Every window has rank at most $2^n \rho_{n,w}$, where

$$\rho_{n,w} = \min\{1, 4w/\sqrt{6\pi(n-1)} + 2K_{\mathrm{BE}}/\sqrt{n}\}.$$

Thus fixed w costs $n - \frac{1}{2}\log_2 n + O(1)$ qubits per block. No numerical value for the theorem's constant is invented.

An elementary version avoids that constant. Let $\ell = n - 1$ and $Z = 2H_n - \ell I$. Exact moments are

$$\tau Z = 0, \quad \tau Z^2 = 3\ell, \quad \tau Z^4 = 27\ell^2 - 30\ell + 24 \le 27\ell^2.$$

For the fourth moment a single bond contributes 21, a disjoint pair 54 after its six orderings, and an adjacent pair 30. The remaining terms have a singly occurring extreme bond and zero trace. Hölder gives $\mathbb{E}|Z| \ge \sqrt{\ell}$; mean zero and Cauchy–Schwarz put probability at least $1/48$ in each tail beyond $\pm\sqrt{\ell}/4$. Let $\alpha(z)$ rise linearly from 0 to $\pi/2$ across those endpoints and stay constant outside. The positive complete effects $A_- = \cos\alpha(Z)$, $A_+ = \sin\alpha(Z)$ obey

$$R_n := \max_x \operatorname{rank} A_x \le \lfloor (47/48) 2^n \rfloor, \quad d \le \delta_{\mathrm{bin}} := \min\{1, 4\pi^2 s^4 (1 + r^{n-1})/(n-1)\}.$$

Their vector derivative norm relative to Z is $\pi/\sqrt{\ell}$, which proves the loss bound from the charge identity.

9.2 Physical readout and exact finite-count error

For successive independent blocks, rotational scalar Gram matrices and zero new-block logical trace give, for arbitrary old correlations,

$$a_{j+1} = a_j + \beta q_j, \quad q_{j+1} = h q_j, \quad 0 \le h \le r^n, \quad 0 \le \beta \le 1 - r^n.$$

Hence $a_K = \beta(1 - h^K)/(1 - h)$. If V_x encodes the support, its branch reference is $V_x A_x^2 V_x^\dagger / 2^n$. The branch Petz map is exactly $Y \mapsto V_x^\dagger Y V_x$, since the A_x factors cancel on support. Unpacking, conditional embedding and reverse inverse collisions with a mixed dummy give a deterministic physical decoder. There is no inverse filter.

Set $\bar{f} = \min\{1, \delta/(1 - r^n)\}$. For every N, retaining any final incomplete block raw, the recovered channel is depolarizing and

$$\epsilon_P \le \tfrac{3}{4}\{r^N + (1 - r^N)\bar{f}\}. \tag{23}$$

This follows from $\beta \ge (1 - \bar{f})(1 - h)$ and $q \le r^N$. For $n s^4 \le 1$, one has $\bar{f}_{\mathrm{grid}} \le \pi^2/(nw^2)$ and $\bar{f}_{\mathrm{bin}} \le 16\pi^2/[n(n-1)]$, capped by one. Target $0 < \eta < 3/4$ is attained if $n(n-1) \ge 24\pi^2/\eta$, $n \sin^4\theta \le 1$ and $r^N \le 2\eta/3$. At arbitrary angle, $n - 1 \ge 12\pi^2/\eta$ suffices instead, but its extra $O(\eta^{-1})$ block-rounding horizon can be costly.

9.3 Integer memory, service and control

Since $(47/48)^{33} < 1/2$, any 33 binary blocks fit into $33n - 1$ qubits, conditional on their classical labels. For N divisible by $33n$,

$$M_{\mathrm{data}} \le N - N/(33n), \qquad M_{\mathrm{core}} \le N - N/(33n) + O(n).$$

Collection, unpacking, flags and the dummy are in the workspace term. General groups of B blocks use $\lceil B \log_2 R \rceil$ packed qubits and Bn raw workspace plus flags; incomplete groups and tails count. Diagonalize the known H_n, attach its coarse flag, measure only the flag, and conditionally relabel its support. Generic synthesis costs $2^{O(B(n+\log v))}$ addressed gates per group with v outcomes. Total work is the number of groups times that cost, plus N inverse gates. A discrete universal gate set adds a polylogarithmic precision overhead; total channel-approximation budget ν adds at most ν to recovery error.

For service time τ_{enc} and arrival spacing Δ, provision $\max\{1, \lceil \tau_{\mathrm{enc}}/(Bn\Delta) \rceil\}$ processors or the corresponding coherent queue, adding all their workspace. Preparation, routing, reset and protection are charged. The core formula is not the total peak at an arbitrary clock; no efficient Bethe-basis transform is assumed.

9.4 A strict comparison that includes extra acquisition

Fix $e = 4\epsilon/3 \in (0, 1)$, $L = \log(1/e)$ and $p_n = 16\pi^2/[n(n-1)]$. Choose fixed n with $p_n < e/2$ and $n > 4224\pi^2/(eL)$. At sufficiently small angle $ns^4 \le 1$. Taking $r^N \le (e - p_n)/(1 - p_n)$, with N rounded up to $33n$, gives

$$\limsup_{\theta \to 0} \sin^4 \theta \, M_{\mathrm{core}} \le \left(1 - \frac{1}{33n}\right) \log \frac{1 - p_n}{e - p_n} \le L - \frac{L}{66n} < L.$$

The estimate is $\log[(1-p_n)/(e-p_n)] \le L + 64\pi^2/(en^2)$. The simple all-output inverse reader has coefficient L. For finite control choose ideal error $\epsilon - \nu(\theta)$ with $\nu(\theta) \to 0$; continuity preserves the strict limiting improvement. At fixed block size and fixed primitive-gate/arrival-time ratio, the needed precision gives polylogarithmic service queues, hence $o(N)$ additional cells. Gate work remains exponential in n, and every protection resource must still be charged. This is a small coefficient improvement over one explicit benchmark, retaining quartic scaling. It is not an optimal law.

10 Why logarithmic-memory Schur sampling does not recover this stream

The tracial central limit theorem [11] makes fixed polynomials of normalized collective sums converge in distribution to classical Gaussian quantities. The physical streaming weak-Schur algorithm [12] uses logarithmic quantum memory. Neither conclusion is a recovery theorem. The operational constructor analyzed the actual latter instrument, granting it every classical path label and its final quantum spin register.

For a block of $b \ge 2$ carriers let

$$x = \sin^2 \theta, \quad B_b = 1 - (1 - x^2)^b, \quad g_b = \tfrac{1}{2} b(b-1)(1-x)^{2b-2} x^2.$$

Every independently processed standard real-Clebsch–Gordan Schur block loses at least $g_b q$ of the incoming residual quantity q. Consequently, after K blocks, even with an unrestricted final

joint decoder,

$$a_K \le \frac{B_b}{B_b + g_b}\left[1 - \{(1-x^2)^b - g_b\}^K\right] \le \frac{B_b}{B_b + g_b}. \tag{24}$$

This is exact at finite b, θ, with arbitrary old quantum coefficients.

Proof. Let P_α be the projector onto the single spin copy specified by a complete standard coupling path. Retaining that copy and the path label is CPTP, and its tracial Petz-back composition is the Hilbert–Schmidt orthogonal projection $\mathcal{E}_b(X) = \sum_\alpha P_\alpha X P_\alpha$. Each path projector is invariant under spin time reversal. The vector $Y_a = \sum_{i<j}(\boldsymbol{\sigma}_i \times \boldsymbol{\sigma}_j)_a$ is time-reversal even, whereas J_a is odd. Within one irreducible spin copy, a Hermitian vector operator is a real multiple of the spin generator. It follows that $P_\alpha Y_a P_\alpha = 0$ and $\mathcal{E}_b(Y_a) = 0$.

One source collision acts exactly as

$$\sigma_S^a I_E \mapsto c^2 \sigma_S^a I_E + s^2 I_S \sigma_E^a - cs(\boldsymbol{\sigma}_S \times \boldsymbol{\sigma}_E)_a.$$

Once the source-identity branch occurs it cannot return to a source-traceless term. Thus the bath-weight-two part of the latter is exactly

$$c^{2b-2}s^2 \sum_{i<j}\{\sigma_j^a(\boldsymbol{\sigma}_S \cdot \boldsymbol{\sigma}_i) - \sigma_S^a(\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j)\}.$$

Its part odd under reversal of bath order is $-c^{2b-2}s^2 Z_a/2$, where $Z_a = (\boldsymbol{\sigma}_S \times \mathbf{Y})_a$. The even part is orthogonal by reversal parity; all other bath weights are orthogonal by Pauli-string support. Source-identity terms are orthogonal on S. Therefore none can cancel this component.

Normalized Pauli orthogonality gives

$$\langle Z_a, Z_d\rangle_{\tau_S \otimes \tau_b} = 2b(b-1)\delta_{ad}.$$

For arbitrary old balanced coefficients B_a, the erased component of $\sum_a \sigma_S^a \otimes B_a$ has squared norm

$$\tfrac{1}{4}c^{4b-4}s^4 \sum_{a,d}\langle Z_a, Z_d\rangle\langle B_a, B_d\rangle = g_b \sum_a \|B_a\|_2^2 .$$

No orthogonality or commutativity of the old coefficients was assumed. The Z_a lie in the pinching kernel, proving the claimed loss. Each block can acquire at most $B_b q$ and leaves at most $\{(1-x^2)^b - g_b\}q$; summing the geometric series proves (24). □

The growing-block limit is controlled by exact inequalities:

$$\frac{b-1}{2}(1-x)^{2b-2} \le \frac{g_b}{B_b} \le \frac{b-1}{2}\left(\frac{1-x}{1+x}\right)^{b-1} \le \frac{b-1}{2}.$$

Hence $B_b/(B_b + g_b) = 2(1 + O(bx))/(b+1)$ uniformly as $bx \to 0$. Choosing $b \to \infty$ with $b\theta^2 \to 0$ makes this reader lose its accessible information. Its unavoidable block loss is of order $b^2\theta^4$, already larger than the $b\theta^4$ acquisition scale.

The ordered quantity has $\|Y_a/b\|_{2,\tau_b}^2 = (b-1)/b \to 1$ and is orthogonal to every polynomial in collective J: bath reversal fixes each J_a and negates Y_a. The required information is therefore absent from a collective-spin sufficient-statistic claim. The exact residual bath-weight distribution is

$$q_{b,k} = \binom{b}{k} c^{4b-2k} 2^k s^{2k}.$$

It also shows that discarding every weight above two is not justified just by $b\theta^2 \ll 1$: the weight-three norm squared divided by acquisition is of order $b^2\theta^2$. Such an operator truncation would need the stronger scale $b\theta \to 0$, and is not itself a CPTP compression. This theorem leaves coherent multiplicity-preserving, old-memory-assisted instruments in the full question; the three-carrier operation above is outside its restricted premise.

11 A recursive spin representation and its physical obstruction

The three-carrier cut exposes an exact representation change. On its rank-two branch, J becomes a Pauli vector and Y vanishes. Identify the rank-six branch with spin one A times spin half B, with $J_A = 2S_A$ and $J_B = \boldsymbol{\sigma}_B$. Exact real intertwiners give

$$J \mapsto J_A + J_B, \qquad Y \mapsto \frac{2}{\sqrt{3}}(J_A \times J_B).$$

This is a genuine compressed-generator identity. It is not ordinary unitary Lax closure. For example, compressing the three-site monodromy to the rank-two doublet gives $M_P(u) = \alpha(u)I + \beta(u)\,\mathrm{SWAP}$, where

$$\alpha = \frac{6u^3 + 6iu^2 + 3u - i}{6}, \quad \beta = \frac{i(6u^2 + 6iu + 5)}{6}, \quad \frac{i\alpha}{\beta} = u - \frac{2u + i}{6u^2 + 6iu + 5}.$$

The last expression is not an affine evaluation shift. Direct exact RTT substitution also gives a nonzero defect for both compressed branches.

More fundamentally, a measured outgoing sector does not prepare that sector before the fixed source interactions. Its actual conditional channel sums over all unobserved incoming bath sectors. The rank-two branch's channel from logical input to source plus retained doublet has Choi rank eight at every finite $u = \cot\theta > 0$. A unitary collision with an independently tracial qubit gauge has Choi rank at most two. The accompanying symbolic amplitude determinant proves that rank, so the extra noise cannot be removed by relabeling the compressed block as a fresh site.

A real fusion tree does close at the first two generators, in the form $J = J_A + \boldsymbol{\sigma}_B$, $Y = \gamma(J_A \times \boldsymbol{\sigma}_B)$. Its complete recursion, branch probabilities and exact fourth-order loss increments are in the algebraic proof file. The following stronger witness shows why this particular hierarchy cannot attain arbitrary accuracy.

11.1 Explicit hypotheses on the instrument effects

At a prefix of $n \geq 3$ carriers, refine the complete instrument and put $A_x = \sqrt{K_x^\dagger K_x}$, $\sum_x A_x^2 = I$. Assume each A_x commutes with collective SU(2), is invariant under spin time reversal, and in $\bigoplus_j (V_j \otimes M_j)$ has the form $\bigoplus_j (I_{V_j} \otimes a_{x,j})$ with each positive $a_{x,j}$ of rank at most one. Real projective fusion trees retaining a payload qubit times one spin gauge satisfy these premises. Generic covariant CPTP maps do not: Kraus tensor operators can exchange spin with discarded ancillas. Complex multiplicity effects and higher retained multiplicities remain in the unrestricted question.

Define scalar chirality

$$C = J \cdot Y = \sum_{i<j<k} \boldsymbol{\sigma}_i \cdot (\boldsymbol{\sigma}_j \times \boldsymbol{\sigma}_k).$$

It is SU(2) invariant and time-reversal odd. In a real Schur basis it is imaginary antisymmetric on each multiplicity space. Thus $A_x C A_x = 0$. The within-spin part of Y is exactly $Y_{\parallel,a} = J_a (J^2)^+ C$, where the inverse vanishes on spin zero. The balanced refined-instrument map obeys $V^\dagger V = \Lambda$, $\Lambda(X) = \sum_x A_x X A_x$. It is self-adjoint, contractive and annihilates every $Y_{\parallel,a}$.

Writing $\tau = 2^{-n}\,\mathrm{Tr}$, Pauli overlap counts give the exact moments

$$\tau(C^2) = n(n-1)(n-2), \qquad \tau(C^2 J^2) = n(n-1)(n-2)(5n-12).$$

For the second identity, the additional pair overlap is $\sum_{i<j}\tau(C^2\boldsymbol{\sigma}_i\cdot\boldsymbol{\sigma}_j) = -18\binom{n}{3} + 24\binom{n}{4}$. Cauchy–Schwarz consequently gives

$$K_n := \sum_a \left\|Y_{\parallel,a}\right\|_{2,\tau}^2 = \tau[C^2(J^2)^+] \geq \frac{n(n-1)(n-2)}{5n-12}.$$

It follows that the positive instrument's leading Yangian energy is at least $K_n/8 \sim n^2/40$. This coefficient statement alone would not justify a growing-prefix recovery obstruction; a finite-angle argument is needed.

11.2 A finite-angle bound that survives the growing prefix

Use the exact Pauli-weight decomposition above and the kernel witnesses $F_a = (\boldsymbol{\sigma}_S \times Y_\parallel)_a$. Their Gram matrix is $2K_n\delta_{ab}/3$. Reversal parity isolates the weight-two overlap $-c^{2n-2}s^2F_a/2$. Bath time reversal excludes odd bath weights from this overlap. Only even weights four and above can contaminate it; their total squared norm is exactly

$$T_{\mathrm{even}}(n,\theta) = \sum_{\substack{4\leq k\leq n\\ k \text{ even}}} \binom{n}{k} c^{4n-2k}2^k s^{2k}.$$

Projection and the triangle inequality give the nonnegative loss bound

$$d_n(\theta) = \left[c^{2n-2}s^2\sqrt{\frac{n(n-1)(n-2)}{6(5n-12)}} - \sqrt{T_{\mathrm{even}}(n,\theta)}\right]_+^2. \tag{25}$$

Because $V^\dagger V = \Lambda$ and $\Lambda(F_a) = 0$, the source-granted Petz visibility after this prefix is at most $1 - d_n$. Arbitrary subsequent processing and additional carriers therefore still obey

$$\epsilon \geq \tfrac{3}{4}\{1 - \sqrt{1 - d_n(\theta)}\}.$$

This holds at every finite angle and integer $n \geq 3$ under the explicit effect hypotheses. It can be vacuous; no perturbative extrapolation is used.

For $n = \lfloor \alpha/\sin^2\theta \rfloor$, the exact bound tends to

$$d(\alpha) = e^{-2\alpha}\left[\frac{\alpha}{\sqrt{30}} - \sqrt{\cosh(2\alpha) - 1 - 2\alpha^2}\right]_+^2.$$

At $\alpha = 0.1$, $d = 0.00008330398585\ldots$, yielding $\liminf \epsilon \geq 0.00003123964530\ldots$. Thus the real multiplicity-free hierarchy has a nonzero weak-angle accuracy floor even though its quantum width grows with the spin. The bound does not settle any fixed target above that floor, and does not extend to arbitrary permitted instruments. A successful replacement must preserve sufficient coherent multiplicity and chirality and prove its cumulative loss, not merely close the collective-spin labels.

12 Initialized coherent frames reveal a different weak-angle limit

Keep k receiver qubits in an independently prepared state. After every source collision, pass the carrier through those k qubits using the same positive-angle partial SWAP, then discard it. This is a physical causal receiver with $k+1$ live qubits during an encounter and k receiver partial-SWAP gates per carrier before readout. A generic final qubit decoder can use a Stinespring

implementation with at most $k+2$ simultaneous qubits, before any additional service or protection workspace. Generic entangled-frame preparation and generic final decoding may be exponentially expensive in k; the retained width does not make their synthesis or service free.

Let $n = k + 1$ count source plus receiver spins and

$$\mathcal{T}_{n,\theta}(X) = \tfrac{1}{2}\operatorname{Tr}_E W_n(\theta)(I_E \otimes X)W_n(\theta)^\dagger, \qquad W_n = e^{i\theta P_{E,n}} \cdots e^{i\theta P_{E,1}}.$$

The source is the first site and is never controlled by the receiver. The channel constructor established, for each *fixed* n and $t > 0$,

$$\mathcal{T}_{n,\theta}^{\lfloor t/\theta^4 \rfloor} \longrightarrow e^{t\mathcal{G}_n}\mathcal{P}_{\mathrm{inv}}, \qquad \mathcal{G}_n(X) = -\tfrac{1}{32}\sum_a [Y_a,[Y_a,X]] + \tfrac{i}{8}[C_n, X], \tag{26}$$

in diamond norm, uniformly on compact positive slow-time intervals. $\mathcal{P}_{\mathrm{inv}}$ is collective SU(2) twirling and $C_n = J \cdot Y$ is scalar chirality. The generator is Lindblad, with Hermitian jumps $Y_a/4$ and Hamiltonian $-C_n/8$. The collective twirl emerges dynamically; it is not an operation requiring access to the source. Relational multiplicity encoding under collective rotations is established reference-frame theory [10]; the ordered-collision generator is the present deduction.

12.1 Proof and the nonuniformity that matters

Write the local analytic logarithm as $\log W_n = i\sum_{r\geq 1}\theta^r H_r$. BCH gives

$$H_1 = \tfrac{n}{2}I + \tfrac{1}{2}\boldsymbol{\sigma}_E \cdot J, \quad H_2 = \tfrac{1}{4}\boldsymbol{\sigma}_E \cdot Y, \quad E_E(H_3) = \frac{J^2 - 3nI}{12}, \quad E_E(H_4) = C_n/8,$$

where $E_E = \operatorname{Tr}_E/2$. The last two identities hold for all n. An order-r BCH term is a homogeneous Lie polynomial supported on at most r sites and has spin-time-reversal parity $(-1)^{r+1}$. At fourth order, conditional trace leaves only a time-reversal-odd SU(2) scalar. Two-site terms and four-distinct-site terms have the wrong parity; the only surviving three-site scalar is chirality, with a universal ordered-triple coefficient. Exact rational SWAP multiplication on three sites gives $1/8$. The same support argument and finite calculation give the third-order dot-product coefficient $1/6$. Both finite identities are included as symbolic certificates.

For X in the full SU(2) commutant, $[H_1, I_E \otimes X] = 0$. If Z is globally invariant and $Z = I_E \otimes Z_0 + \sum_a \sigma_E^a \otimes Z_a$, the Z_a transform as a vector and $E_E[H_1, Z] = \frac{1}{2}\sum_a [J_a, Z_a] = 0$. All nested terms with an outer H_1 therefore vanish after trace, and those with an inner H_1 vanish on X. The third-order trace commutes with X. The surviving fourth-order terms are

$$i[E_E H_4, X] - \tfrac{1}{2}E_E[H_2,[H_2, I_E \otimes X]],$$

which prove the displayed $\mathcal{G}_n$ with the stated Schrödinger sign.

Exact SU(2) equivariance keeps the invariant and nonscalar sectors separate. On an operator-spin-$l \geq 1$ sector the channel is $I - \frac{1}{2}l(l+1)\theta^2 + O_n(|\theta|^3)$, so its repeated norm decays exponentially in t/θ^2. On the invariant sector the expansion is $I + \theta^4\mathcal{G}_n + O_n(|\theta|^5)$ and the exponential-product formula applies. Fixed-dimensional norm equivalence gives diamond convergence. This proves (26), including arbitrary reference inputs. The constants depend on n; no exchange of the width and angle limits is justified by this theorem.

12.2 An explicit finite-angle and finite-count estimate

The root converted that fixed-width statement into the following conservative uniform estimate, independently checked by the channel constructor. Put

$$R_p = e^{2n\theta}(2n\theta)^p/p!, \quad \Gamma_n = \tfrac{3}{8}n^2(n-1)^2 + \tfrac{1}{4}n(n-1)(n-2),$$

$$\eta = 1 - \theta^2 + R_3, \qquad E = R_5 + \tfrac{1}{2}\theta^8 \Gamma_n^2 e^{\theta^4 \Gamma_n}.$$

If $0 < \theta \le 1$, $\theta^2(1 + n(n+1)/2) \le 2$, and $\eta < 1$, then for every integer $N \ge 1$,

$$\tfrac{1}{2} \left\| \mathcal{T}_{n,\theta}^N - e^{N\theta^4 \mathcal{G}_n} \mathcal{P}_{\mathrm{inv}} \right\|_\diamond \le \tfrac{1}{2}\{NE + 2^n \eta^N\}. \tag{27}$$

Comparison with a chosen t adds at most $\Gamma_n |t - N\theta^4|/2$. The bound also holds after any initialized frame and physical final readout. It does not require access to the actual source.

For proof, each conjugation generator has diamond and Hilbert–Schmidt operator norms at most 2. The multinomial theorem bounds the degree-r coefficient of the ordered product by $(2n)^r/r!$. Injection of $I_E/2$ and partial trace contract diamond norm; their Hilbert–Schmidt norms $1/\sqrt{2}$ and $\sqrt{2}$ cancel. The scalar remainder is at most R_5. Since $\|Y_a\| \le n(n-1)$ and $\|C_n\| \le n(n-1)(n-2)$, the generator's diamond norm is at most Γ_n. The exponential remainder supplies the second term of E; CPTP telescoping gives NE. On nonscalar sectors the second-order coefficient has eigenvalues $-l(l+1)/2$, $1 \le l \le n$. The angle condition and third-order remainder give contraction η. Finally $\|\Lambda\|_\diamond \le 2^n \|\Lambda\|_{2\to 2}$ yields the transient term. This proves (27).

The sufficient small-angle condition scales at worst as $\theta = O(n^{-3})$, and the leading scalar error at fixed t is $O(tn^5\theta)$. These estimates are explicit, not sharp. For G_{phys} elementary operations, individual half-diamond error at most δ/G_{phys} is a sufficient control budget. In particular receiver-angle error at most $\delta/(kN)$ suffices for the kN partial SWAPs. The actual service must fit k encounters per carrier or pay for additional coherent queues. Preparation, readout and protection of all live qubits remain part of the execution cost.

12.3 A rigorous counterexample to vanishing accessible information

Use two receiver qubits initialized in $|0\rangle|+\rangle$. Writing $D_{12} = \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$, the exact limiting accessible channel is

$$\mathcal{E}_t((I + \mathbf{r} \cdot \boldsymbol{\sigma})/2) = \frac{I_4}{4} + \frac{D_{12}}{12}\{u_x(t) r_x + u_z(t) r_z\},$$

where

$$u_x = -\tfrac{1}{2}(e^{-t} - e^{-3t}), \qquad u_z = e^{-t}\left(\tfrac{3}{2}t - \tfrac{1}{4} + \tfrac{1}{4}e^{-2t}\right).$$

Consequently $a(t) = (u_x^2 + u_z^2)/9 > 0$ for every $t > 0$; $a(1)$ is about 0.02759547. For a short exact witness, the receiver singlet projector $P_- = (I - D_{12})/4$ obeys

$$\mathcal{G}_3^*(I_S \otimes P_-) = \tfrac{1}{4}(-D_{S1} + D_{S2} + D_{12}), \qquad p_-'(0) = \tfrac{1}{4}(r_x - r_z).$$

Thus opposite source Z inputs have different receiver probabilities at small positive slow time. This refutes a proposed universal bound $a_N \le C(m)\theta^2$ at fixed memory dimension, uniformly in N. The two-qubit example's output algebra is abelian, so its information is classical and the channel is entanglement breaking. It is not accurate reference-preserving recovery.

Optimization over pure initial frames and physical final decoders gave the following feasible values at slow time $t = 1$:

Retained qubits k	3	4	6	8	9
Bell fidelity	0.384710	0.414232	0.461009	0.495810	0.510259

These are local numerical optima of the specified static cascade, not upper bounds on any receiver class. The saved initial states and repaired CPTP decoder matrices specify feasible operations.

Separate stage-control and finite-feedback searches are also included. None of the completed families proves high-fidelity recovery as width grows.

The $k = 9$ value exceeds the 1/2 Bell-fidelity ceiling of an entanglement-breaking qubit channel. Independent explicit Choi replay gives 0.510258885281022, negative partial-transpose eigenvalue -0.01270409, and trace-preservation residual below 1.4×10^{-13}. Splitting the semigroup calculation into seven pieces changes the recovered Choi state by trace distance below 6×10^{-16}. Positive-part and marginal normalization of the saved matrices specify an exact CPTP readout. These are floating-point checks, not an interval-certified fidelity proof. They support fixed-width entanglement transmission as a construction opening; they establish neither high-accuracy scaling nor positive unassisted quantum capacity. Equation (27) supplies exact angle quantifiers for any separately certified limiting fidelity.

12.4 A proved all-width reference-frame boundary

For the prescribed passive matched cascade with any initialized frame and any final decoder, its quartic-time limit satisfies

$$\epsilon_{k,t} \geq h_k := \frac{1 - \cos(\pi/(k+2))}{2}, \qquad t > 0.$$

The channel constructor proved this by granting the actual source immediately after the initial collective twirl, then relaxing SU(2) to one-axis U(1) twirling. Full twirling is a postprocessing of the latter. For a pure k-qubit frame with charge probabilities $p_0, \ldots, p_k$, the total-charge blocks of a Bell input have two Schmidt coefficients $\sqrt{p_h/2}, \sqrt{p_{h-1}/2}$, with outside probabilities zero. Their optimal Bell overlaps sum to

$$F_{U(1)}(p) = \tfrac{1}{2} + \tfrac{1}{2} \sum_{h=0}^{k-1} \sqrt{p_h p_{h+1}}.$$

Writing $a_h = \sqrt{p_h}$ reduces the optimization to the path adjacency matrix divided by two. Its top eigenvalue is $\cos(\pi/(k+2))$, with sine eigenvector $a_h = \sqrt{2/(k+2)}\sin[(h+1)\pi/(k+2)]$. Mixed frames and independent classical labels cannot improve this maximum. The sine optimization is established phase-reference machinery [16,17]. Its attainment is for the source-granted U(1) task; it does not decode the actual source-erased SU(2) stream.

Let $\Delta_{k+1}(\theta, N)$ denote the right side of (27). Under that estimate's stated hypotheses, the actual finite-angle cascade satisfies

$$\epsilon_{\text{physical}}(k, \theta, N) \geq [h_k - \Delta_{k+1}(\theta, N)]_+.$$

This is uniform over angle-dependent initializations and final decoders. One fully joint consequence is: for fixed $0 < A \leq B < \infty$ and $L = \log(1/\epsilon)$, if $AL \leq N\theta^4 \leq BL$ and

$$\theta = o(\epsilon^{7/2}/L), \quad \text{then} \quad k + 2 \geq (\pi/2 + o(1))\epsilon^{-1/2}.$$

For $k = O(\epsilon^{-1/2})$ the scalar remainder is $O(Lk^5\theta) = o(\epsilon)$, the next remainder is $O(L\theta^4 k^8) = o(\epsilon)$ and the transient vanishes uniformly; larger widths already satisfy the bound. The angle exponent is a conservative sufficient regime, not a sharp transition. This architecture therefore pays at least order $\theta^{-4}\epsilon^{-1/2}\log(1/\epsilon)$ receiver encounters there. The result does not restrict arbitrary causal instruments or prove a matching family of passive frames and decoders.

12.5 Directed continuity and its remaining quantum requirement

Exact circuit causality gives $\mathcal{T}^*_{n,\theta}(X_{[1,j]} \otimes I) = \mathcal{T}^*_{j,\theta}(X_{[1,j]}) \otimes I$. For invariant X, the same identity holds for $\mathcal{G}^*$. Thus the prefix invariant algebras form a triangular flag, with no inference from ordinary eigenvalues to singular contraction.

The operational constructor proved an exact continuity law. Set $D_j = \boldsymbol{\sigma}_j \cdot \boldsymbol{\sigma}_{j+1}$, $F_j = J_{[1,j]} \cdot \boldsymbol{\sigma}_{j+1}$, and $F_0 = 0$. Then

$$\mathcal{G}^*_n D_j = F_{j-1} - F_j, \qquad \mathcal{G}^*_n \sum_{j=1}^{n-1} D_j = -F_{n-1}.$$

For an adjacent swap P_j, one has $P_j Y_a P_j = Y_a - 2(\boldsymbol{\sigma}_j \times \boldsymbol{\sigma}_{j+1})_a$. Substitution into the dissipator shows that its downstream terms cancel exactly against the chirality commutator, leaving the displayed upstream current. The full derivation and exact Pauli-string certificates are included.

For $H_w = \sum_j \sin(\pi j/n) D_j$, Pauli orthogonality further gives

$$\frac{-\langle H_w, \mathcal{G}^*_n H_w\rangle}{\|H_w\|_2^2} = 2\sin^2\frac{\pi}{2n}, \qquad \frac{\left\|e^{t\mathcal{G}^*_n} H_w - H_w\right\|_2}{\|H_w\|_2} \le \frac{\pi t}{\sqrt{n}}.$$

The last bound follows by integrating $\mathcal{G}^*_n H_w$ and using Hilbert–Schmidt contraction. This is an all-width slow-observable theorem. A single protected Hermitian observable can carry classical information; an extractable qubit requires three compatible noncommuting logical observables and a physical decoder. Moreover, the pair-observable sector already fails to close on four sites. The continuity law therefore does not turn the full dynamics into a classical nearest-neighbor transport model.

12.6 Why an existing coherent absorber cannot simply be substituted

The general coherent-reversal construction [13] uses a pure traveling field state and a source–absorber purification of the stationary source. Recent absorber estimation protocols [14] likewise use a pure joint stationary state. For the present tracial stream, however, every permitted independently initialized receiver obeys the exact average identity $\mathcal{N}_j(I/2) = I_S/2 \otimes \sigma_j$, also in each classical branch. A purification $|\psi\rangle_{SM}$ of $I_S/2$ satisfies

$$\langle\psi|(I_S/2 \otimes \sigma_j)|\psi\rangle = \tfrac{1}{4}\operatorname{Tr}(P_{\psi,M}\sigma_j) \le \tfrac{1}{4}.$$

Thus that pure source–absorber stationary state cannot be reached from the allowed initialization. Purifying each tracial carrier formally would enlarge the traveling field by its inaccessible purifier. The cited unitary receiver on that enlarged field is not an allowed operation here. This is a precise failure of those premises, not a no-go for transient recovery into the actual receiver or its inaccessible logical reference.

13 An exact finite policy boundary with physical attainment

The channel and operational constructors independently derived the same finite dynamic programme. Its unnormalized planning state ρ_{RSM} includes R, S only for offline synthesis, not as receiver controls. For $K : ME \to M$, let

$$\Xi_K(\rho) = (I_{RS} \otimes K) U_{SE}(\rho \otimes \tau_E) U_{SE}^\dagger (I_{RS} \otimes K^\dagger).$$

For retained dimension m and arbitrary classical histories define

$$V_0(\rho) = \tfrac{1}{2}\min\{\operatorname{Tr} Z_M : I_R \otimes Z_M \geq \operatorname{Tr}_S \rho\}, \tag{28}$$

$$V_t(\rho) = \max_{\sum_x K_x^\dagger K_x = I} \sum_x V_{t-1}(\Xi_{K_x}(\rho)). \tag{29}$$

The terminal expression is the standard min-entropy/singlet-fraction SDP [8]. The exact optimum half-diamond error is

$$\epsilon_m^*(N,\theta) = 1 - V_N(\Phi_{RS} \otimes |0\rangle\langle 0|_M).$$

Convexity makes some pure initialization at least as good as any mixed one; all pure initializations are equivalent under allowed controls. A global Clifford twirl converts optimum Bell fidelity into optimum half-diamond error without adding retained quantum memory.

Normalize each nonzero K to Frobenius norm one and place its squared norm in a nonnegative weight. This gives a compact measure problem with a Hermitian $2m \times 2m$ moment constraint. Conic Carathéodory on that moment and the objective yields an attaining instrument with at most $4m^2 + 1$ outcomes per node. Induction gives an attaining finite tree. This also removes any supremum loophole from continuous classical outcomes. For algebraic c, s, a finite threshold is decidable in principle by real quantifier elimination; neither efficient synthesis nor useful asymptotic scaling follows.

A local dual is $V_t(\rho) = \inf_H \operatorname{Tr} H$ subject to

$$\operatorname{Tr}(HK^\dagger K) \geq V_{t-1}(\Xi_K(\rho)) \quad \text{for every Frobenius-unit } K.$$

It provides compatible converse certificates through actual reachable states, whereas unrelated good cut compressions need not compose. This compatibility issue is established quantum-comb memory theory [9].

A sequential binary-POVM implementation uses the retained register, current carrier and one resettable pointer: at most $\lceil \log_2 m \rceil + 2$ coherent qubits after padding. A crude generic synthesis uses $O(Nm^4)$ two-qubit gates before finite-gate-set precision overhead, with a potentially exponential classical policy table. Slower service than arrivals creates a quantum queue. This is an attained finite boundary with an execution bill, but its unknown parameter dependence is precisely why it is not the requested solved law.

14 A physical benchmark with paid routing and control

On a line of stored carriers, a neighboring gate

$$W_\theta = \text{SWAP } U_\theta^\dagger = c\,\text{SWAP} - isI = -iU_{\pi/2-\theta}$$

both reverses one collision and moves the dummy one cell left. For K stored carriers, the newest-to-oldest sweep uses $K+1$ qubits and K neighboring gates with exactly error $3r^K/4$. Its angle approaches $\pi/2$ as $\theta \to 0$; the service time does not become weak automatically.

A specified fixed-input-port ladder has K data cells and K reset routing cells. Two parallel swap layers shift the data between arrivals and restore the routing row. This bills $2K+1$ resident coherent cells, any transient incoming carrier, $K(K-1)$ loading routing swaps, K loads and K final sweep gates. The quadratic loading count belongs to this architecture, not to every implementation. For arrival spacing Δ and final gate time τ_g, the protected qubit-time of stored carriers and the decoding dummy is

$$V = \Delta K(K-1)/2 + \tau_g K(K+3)/2.$$

Preparation, reset, angle calibration, control bits and protection of all additional routing and work cells must be added. A channel hybrid bound allocates gate, preparation and storage errors. No unknown physical noise rate or free protection is substituted for a recovery construction. Any service backlog is coherent memory.

15 Why normalization alone does not close the crossing gap

The operational constructor tested whether scalar normalization of the preserved crossing argument could upgrade its singular-gap bound from $\exp[-O_\theta(n \log n)]$ to $\exp[-O_\theta(n)]$. For $u = \cot\theta$, $a_j = \sqrt{u^2 + j^2}$, and the degree-n monodromy $M_n(z)$, normalized inverse transfer is contractive, but crossing retains the exact scalar factor

$$\mathcal{C}\left[\frac{M_n(z-i)}{z^n}\right] = \left(\frac{z-i}{z}\right)^n \left(\frac{M_n(z)}{(z+i)^n}\right)^{-1}.$$

Restoring polynomial form for interpolation restores the removed factor. The existing residual bounds have envelopes $W_j = (a_{2j-1}a_{2j}/u)^n$ at $z = u - 2ij$.

There is an explicit scalar degree-n polynomial q_n and $\delta_n > 0$ with $q_n(0) = 1$, $|q_n(u)| = W_0\delta_n$, and

$$|q_n(u - 2ij)| \le W_j\delta_n \quad (j \ge 0), \qquad \log\delta_n = -2n\log n + O_u(n).$$

Start with $p_n(z) = \prod_{l=0}^{n-1}(z-u+2il)/(-u+2il)$, then add an explicit positive constant and rescale; the complete certificate and constants are in the accompanying derivation. Thus polynomial degree, initial normalization and even infinitely many of these norm envelopes cannot imply an exponential-in-n coercivity constant. Exact interpolation weights improve finite constants while keeping $\log K_n = 2n\log n + O_u(n)$. This is a proof-method obstruction. The scalar certificate is not a physical intertwiner or a near-null mode. The stronger matrix compatibility before taking absolute norms remains available; no new physical memory lower bound follows from this certificate.

16 Verification and remaining operation

Check	Result and scope
General ledger: 48 random CPTP cases	Nontracial initialization, recorded or discarded outcomes and feedback; residuals below 3×10^{-15}.
Local preprocessing: 150 dense cases	Exact Gram recurrences through five carriers; maximum residual below 5×10^{-16}.
Weak-reader witness: 60 local cases	Arbitrary receiver-only isometries and coherent flags; no violated norm bound, identities below 8×10^{-16}.
Routed inverse: 16 dense cases	One to four carriers; recovered channel checked, maximum entry residual below 3×10^{-16}.
Schur obstruction: 20 dense cases	Two to six carriers, four angles; no violated lower bound beyond 3×10^{-16} roundoff.
Chirality obstruction: 16 cases	Exact kernel and finite-angle tail; kernel residual below 8×10^{-15}.
Saved policy replay: six policies	Full instrument execution; maximum recovered-state trace-norm drift below 1.1×10^{-13}.
Nine-qubit frame/readout	Explicit Choi action and separate semigroup replay; Bell fidelity 0.510258885, floating-point audit only.
Energy-window reader: 58 cases	Train and charge-loss identities, complete instruments and two-block recurrence; key residuals below 3×10^{-15}.
BCH, positive three-site block and continuity	Exact symbolic/rational identities, with derivations extending the stated finite checks.

These finite calculations check identities and implementations. Policy searches provide feasible channels or local optima, not global converses. Proof sources, code, diagnostic policies and exact limitations accompany the report. The full endpoint still requires an attainable width-dependent bound on L_N with paid service and control. The fixed-angle logarithmic gap and weak-angle fixed-error memory optimum have not been settled.

Methods, independence and attribution

Three constructors began in separately initialized contexts with the same full target and accepted prior results. One used channel sufficiency and recovery theory; one received the operational input/action/output problem; one used crossing and spin-chain algebra. Initial approaches were not shared until concrete derivations existed. Mechanisms and audits were then exchanged explicitly. They share training and supplied prior machinery; these are distinct construction attempts, not statistically independent blind trials. The root developed the changing-reference ledger and retention bound, then generalized the algebraic environmental obstruction.

The following primary sources supply existing machinery, not a published solution of this source-specific endpoint. New statements here are research derivations checked within this run, not externally peer reviewed.

1. K. Temme, M. J. Kastoryano, M. B. Ruskai, M. M. Wolf and F. Verstraete, *The χ^2-divergence and mixing times of quantum Markov processes*, J. Math. Phys. 51, 122201 (2010), Theorem 4. https://arxiv.org/abs/1005.2358.

2. S. Beigi, *Sandwiched Rényi divergence satisfies data processing inequality*, J. Math. Phys. 54, 122202 (2013). https://arxiv.org/abs/1306.5920.

3. R. L. Frank and E. H. Lieb, *Monotonicity of a relative Rényi entropy*, J. Math. Phys. 54, 122201 (2013). https://arxiv.org/abs/1306.5358.

4. J. M. Renes, *Better bounds on optimal measurement and entanglement recovery, with applications to uncertainty and monogamy relations*, Phys. Rev. A 96, 042328 (2017). https://arxiv.org/abs/1707.01114.

5. D. Kretschmann, D. Schlingemann and R. F. Werner, *The information-disturbance tradeoff and the continuity of Stinespring's representation*, IEEE Trans. Inf. Theory 54, 1708–1717 (2008). https://arxiv.org/abs/quant-ph/0605009.

6. F. Loebbert, *Lectures on Yangian symmetry* (2016), first realization and coproduct. https://arxiv.org/abs/1606.02947.

7. A. I. Molev, *Irreducibility criterion for tensor products of Yangian evaluation modules*, Duke Math. J. 112 (2002), Theorems 1.1 and 1.2. https://arxiv.org/abs/math/0009183.

8. R. König, R. Renner and C. Schaffner, *The operational meaning of min- and max-entropy*, IEEE Trans. Inf. Theory 55, 4337–4347 (2009). https://doi.org/10.1109/TIT.2009.2025545.

9. A. Bisio, G. M. D'Ariano, P. Perinotti and M. Sedlák, *Memory cost of quantum protocols*, Phys. Rev. A 85, 032333 (2012). https://arxiv.org/abs/1112.3853.

10. S. D. Bartlett, T. Rudolph and R. W. Spekkens, *Classical and quantum communication without a shared reference frame*, Phys. Rev. Lett. 91, 027901 (2003). https://arxiv.org/abs/quant-ph/0302111.

11. G. Kuperberg, *A tracial quantum central limit theorem* (2002), Theorem 1. https://arxiv.org/abs/math-ph/0202035.

12. E. Cervero-Martín and L. Mančinska, *Weak Schur sampling with logarithmic quantum memory* (2023), Algorithm 1 and its sampling guarantee. https://arxiv.org/abs/2309.11947.

13. M. Tsang, *Quantum reversal: a general theory of coherent quantum absorbers*, Quantum 9, 1650 (2025), Sections 2.1–2.2. https://arxiv.org/abs/2402.02502.

14. F. Girotti, A. Godley and M. Guţă, *Estimating quantum Markov chains using coherent absorber post-processing and pattern counting estimator*, Quantum 9, 1835 (2025), coherent-absorber construction. https://arxiv.org/abs/2408.00626.

15. M. Cramer, F. G. S. L. Brandão, M. Guţă, A. M. Alhambra and M. Scandi, *A Berry–Esseen Bound for Quantum Lattice Systems* (2026), Theorem 1, product-state case. https://arxiv.org/abs/2605.03829.

16. D. W. Berry and H. M. Wiseman, *Optimal states and almost optimal adaptive measurements for quantum interferometry*, Phys. Rev. Lett. 85, 5098 (2000), Eqs. 9–10. https://arxiv.org/abs/quant-ph/0009117.

17. S. D. Bartlett, T. Rudolph, R. W. Spekkens and P. S. Turner, *Quantum communication using a bounded-size quantum reference frame*, New J. Phys. 11, 063013 (2009). https://arxiv.org/abs/0812.5040.

18. P. W. Claeys, J. Herzog-Arbeitman and A. Lamacraft, *Correlations and commuting transfer matrices in integrable unitary circuits*, SciPost Phys. 12, 007 (2022), Sections 2.1–2.2. https://arxiv.org/abs/2106.00640.

Astra provenance. [A1] The accepted tracial acquisition proof and prior-art comparison at repository pin

`fef3640a8b1dc5475674a006de417f652f3baf3c`

in https://github.com/harjassand/astra-research-knowledge. [A2] The completed peer general spectral-profile converse and Schur/fusion audits, with their original scope retained, in return

`2026-11-05-p2-weak-stream-memory`

[A3] The preceding causal-recovery proof note, in return

`2026-11-05-p1-causal-recovery`

Retrieved file hashes and source paths are recorded in the accompanying manifest. The original old PDF and archive are preserved byte for byte in the continuation archive.

Preserved earlier proof note

All-prefix crossing bound, refined instrument witness,
exact pre-access carrier-loss law and physical reverse decoder

The following eleven pages reproduce the preceding report without modifying its proof text or attribution. Its page numbering restarts. The original PDF is also included unchanged in the archive. The new results and unresolved joint endpoint are stated in the continuation above.

Causal recovery from a homogeneous weak quantum stream

Quantitative memory lower bounds, exact carrier-loss bounds, and the unresolved scaling question

Research construction and reproducibility note

5 November 2026

Abstract

The fixed endpoint is a causal decoder for an arbitrary qubit emitted through a homogeneous weak partial-SWAP stream, attaining the quartic collision scale with genuinely sublinear quantum storage and near-linear processing, or a matching general obstruction. This investigation does not resolve that endpoint. A crossing and interpolation argument establishes an explicit positive singular gap for every finite prefix. Combined with a strengthened general instrument converse, this forces quantum memory $\Omega(\log(1/\epsilon)/\log\log(1/\epsilon))$ at each fixed nonzero angle, or $\Omega_\theta(N/\log N)$ at the benchmark count. It rules out every fixed polynomial saving in storage in that regime, while leaving a logarithmic gap to the known reader. Exact one- and two-qubit error floors and a carrier-erasure law are also derived. The joint weak-angle limit and the full memory/processing tradeoff remain unresolved.

1 The endpoint remains fixed

One source qubit S, initially in an arbitrary state possibly entangled with an inaccessible reference, meets fresh independent carriers E_j in $\tau = I/2$. Each encounter uses the known gate

$$U_j = cI + is\,\mathrm{SWAP}_{S,E_j}, \qquad c = \cos\theta, \quad s = \sin\theta, \qquad 0 < \theta < \pi/2.$$

The receiver sees each emitted carrier once in chronological order. The final source is discarded. The receiver may use initialized workspace, arbitrary instruments, classical records, feedback and buffers. All coherent registers crossing a time cut count as quantum memory. Bath purifiers, preparation keys and a return link to the source are unavailable.

Write $\mathcal{C}_N$ for the acquired channel and use half-diamond error

$$d_\diamond(\mathcal{D}\mathcal{C}_N, \mathrm{id}) = \tfrac{1}{2}\left\|\mathcal{D}\mathcal{C}_N - \mathrm{id}\right\|_\diamond.$$

The full question asks for a physical causal operation with this error at most ϵ, collision count $N = \Theta(\theta^{-4}\log(1/\epsilon))$, genuinely sublinear retained quantum memory, and near-linear processing, with the resource dependence exposed jointly. A failure of a particular receiver does not answer that question. Nor does a compressed description of an operator establish a physically implementable quantum memory.

Partial-SWAP homogenization itself is established research [1]. General transpose/Petz recovery and its near-optimality are also established machinery [2,3]. The exact source-specific all-output benchmark used here is the earlier Astra return [A1]. The recent general memory converse is [A2]. These are reused, rather than claimed as discoveries in this run.

2 The accepted all-output benchmark

Let $b = s^4$ and $r = 1 - b$. Retain all N outputs, prepare a separate mixed dummy qubit, and apply the inverse collisions to the dummy in reverse chronological order. This is an output-only operation. The actual source is not supplied. Its exact recovered channel is

$$\mathcal{D}_N\mathcal{C}_N = (1 - r^N)\,\mathrm{id} + r^N\mathcal{R}_{I/2}, \qquad d_\diamond(\mathcal{D}_N\mathcal{C}_N, \mathrm{id}) = \tfrac{3}{4}r^N. \tag{1}$$

Every unrestricted decoder of the full output obeys [A1]

$$\epsilon_{\mathrm{opt}} \geq \tfrac{3}{4}\big(1 - \sqrt{1 - r^N}\big). \tag{2}$$

Thus the explicit operation is within a factor two in error of this lower bound. A sufficient count is

$$N_{\mathrm{all}} = \left\lceil \frac{\log(3/(4\epsilon))}{-\log(1 - \sin^4\theta)} \right\rceil. \tag{3}$$

At $\theta = 0.1$ and $\epsilon = 0.01$, it requires 43 462 encounters; the unrestricted converse requires at least 36 552. Source marginal forgetting instead has visibility c^{2N} and therefore a quadratic weak-angle clock. Forgetting and accessible recovery are distinct operations.

The abstract decoder uses N stored carrier qubits, one dummy, and N addressed inverse two-qubit gates. Physical routing, service backlog, protection and calibration must also be counted. Linear addressed gate count is not a geometry-independent linear-time implementation.

3 A strengthened converse for general causal receivers

The recent Astra note [A2] supplies a general spectral-profile reduction. The following construction retains that reduction and improves its finite-error coefficient using established sharp recovery machinery. At prefix length L, let $D = 2^L$ and define the bistochastic bath channel

$$T_L(K) = \tfrac{1}{2}\,\mathrm{Tr}_S\Big[U^{(L)}(I_S \otimes K)U^{(L)\dagger}\Big], \qquad H_L = I - T_L T_L^*.$$

Write its eigenvalues in increasing order as $\lambda_0, \ldots, \lambda_{D^2-1}$. For a converse, grant the receiver the actual source at this cut. It can simulate the future from there, so this only helps recovery.

The previous coefficient in [A2] was $3\epsilon(1 - \epsilon)$. We obtain the following strengthened witness for $0 \leq \epsilon \leq 1/2$.

Theorem 1 (Refined-instrument spectral witness)**.** *Any successful receiver whose quantum capacity in every classical branch at the cut is at most m induces a bistochastic Choi witness Z with*

$$\mathrm{Tr}\, Z = 1, \qquad 0 \leq Z \leq (m/D)I, \qquad \mathrm{Tr}(H_L Z) \leq e_\epsilon := 2\epsilon - \tfrac{4}{3}\epsilon^2.$$

This includes arbitrary initialized workspace, instruments and classical histories. If D/m is an integer, it implies

$$\frac{m}{D}\sum_{j=0}^{D/m-1} \lambda_j \leq e_\epsilon. \tag{4}$$

For every $t > 0$ there is also the heat-trace bound

$$m \geq \frac{D\exp[-te_\epsilon]}{\mathrm{Tr}\exp(-tH_L)}.$$

Proof. Refine each branch of the prefix instrument into its individual Kraus outcomes. Revealing this refinement only helps the receiver. Each Kraus map $K_x : \mathbb{C}^D \to M_x$ has rank at most m. Set

$$A_x = (K_x^\dagger K_x)^{1/2}, \qquad \sum_x A_x^2 = I, \qquad \mathcal{L}(X) = \sum_x A_x X A_x.$$

The polar isometry in K_x is undone by the branch Petz map. Since the logical maximally mixed input is mapped by the reversible prefix encoder to $I_S/2 \otimes I/D$, the logical-channel Petz composition is precisely: encode, apply $\mathrm{id}_S \otimes \mathcal{L}$, undo the prefix unitary and trace the bath. A logical Pauli twirl therefore gives its entanglement fidelity as

$$F_{\mathrm{P}} = \frac{1}{D} \sum_x \|T_L^*(A_x)\|_2^2 .$$

Here the norm is the ordinary, unnormalized Hilbert–Schmidt norm. With ordinary matrix vectorization, define

$$Z = \frac{1}{D} \sum_x |A_x\rangle\!\rangle\langle\!\langle A_x|.$$

Then $\operatorname{Tr} Z = 1$, both marginals are I/D, and exactly

$$\operatorname{Tr}(H_L Z) = 1 - F_{\mathrm{P}}.$$

For an arbitrary matrix B, Cauchy–Schwarz on the support of A_x gives

$$|\operatorname{Tr}(B^\dagger A_x)|^2 \le m \operatorname{Tr}(A_x^2 B B^\dagger).$$

Summing proves $Z \le (m/D) I$.

The sharp qubit Petz bound [2, Eq. (13)] is

$$F_{\mathrm{P}} \ge F^2 + \frac{(1-F)^2}{3}$$

for the entanglement fidelity F of any recovery. Half-diamond error at most ϵ gives $F \ge 1 - \epsilon$. The right side increases for $F \ge 1/4$, so $1 - F_{\mathrm{P}} \le 2\epsilon - 4\epsilon^2/3$ for $\epsilon \le 3/4$.

For completeness, this recovery inequality also follows directly by applying the order-two sandwiched Rényi data-processing inequality to the two-outcome Bell test after recovery. Its success probability on the normalized channel Choi state is F, and on the product of the uniform reference and the channel output average is $1/4$. The resulting binary expression is $4F^2 + (4/3)(1-F)^2$, while the initial expression is $4F_{\mathrm{P}}$.

The density cap minimizes energy by filling the smallest eigenvalues first, proving (4). Jensen's inequality and the same cap give $e^{-te_\epsilon} \le \operatorname{Tr}(Z e^{-tH_L}) \le (m/D) \operatorname{Tr}(e^{-tH_L})$. □

For noninteger D/m, write $a = m/D$ and $k = \lfloor 1/a \rfloor$. The exact profile condition is $a \sum_{j=0}^{k-1} \lambda_j + (1 - ka)\lambda_k \le e_\epsilon$. If $n_L(u)$ counts eigenvalues below $u > 0$, then $m \ge D[1 - e_\epsilon/u]_+ / n_L(u)$. Grouping the refined Kraus terms by the original branch gives, for variable capacity m_y,

$$\Pr(m_y > m) \ge [1 - e_\epsilon/u - m n_L(u)/D]_+.$$

The probability is the branch law under the tracial bath marginal. Summing tail bounds gives expected occupancy bounds as in [A2]. These conditions remain necessary, not sufficient for a compression channel. The new coefficient is strictly smaller than the old one for $0 < \epsilon < 3/5$, including the entire stated range $\epsilon \le 1/2$.

3.1 An explicit gap at every prefix length

The central additional step is a quantitative bound on the singular defect, with no extrapolation from small matrices.

Theorem 2 (All-prefix crossing bound)**.** *For $u = \cot\theta > 0$ and $n \geq 1$, define*

$$B_n = \frac{u^2 + 4n^2}{u}, \qquad K_n = \frac{B_n^n[(u+2n)^n + 1]}{n!}, \qquad g_n = \frac{6}{n(n+1)(2n+1)K_n^2}.$$

Then

$$\gamma_n := 1 - \|T_n|_{\mathbf{1}^\perp}\|_{2\to 2}^2 \geq g_n > 0. \tag{5}$$

The identity is the only zero mode of $I - T_n T_n^*$. At fixed θ, $\log g_n = -4n\log n + O_\theta(n)$.

Proof. Use normalized Hilbert–Schmidt norms unless an operator norm is specified. Define

$$M_n(z) = (zI + iP_{S,n})\cdots(zI + iP_{S,1}), \qquad U^{(n)} = (u^2+1)^{-n/2} M_n(u).$$

For $\mathcal{C}(A) = (\sigma_y \otimes I)A^{t_S}(\sigma_y \otimes I)$, the local identity $\mathcal{C}(P) = I - P$ implies

$$\mathcal{C}[M_n(z-i)] = (z^2+1)^n M_n(z)^{-1}. \tag{6}$$

The bath entries in distinct local factors commute, justifying their reversed order under source transpose. No general multiplicativity of partial transpose is assumed. The map $\mathcal{C}$ preserves Hilbert–Schmidt norm.

Take a traceless bath operator X of norm one, set $Y = T_n(X)$, and write $\widehat{X} = I_S \otimes X$, $\widehat{Y} = I_S \otimes Y$. Source conditional expectation is an orthogonal projection, so

$$\delta^2 := \left\| U^{(n)} \widehat{X} U^{(n)\dagger} - \widehat{Y} \right\|^2 = 1 - \|T_n(X)\|^2 .$$

For $E(z) = \widehat{Y} M_n(z) - M_n(z)\widehat{X}$, one has $\|E(u)\| = (u^2+1)^{n/2}\delta$. If $\widetilde{E}$ interchanges X, Y, crossing gives

$$\widetilde{E}(z-i) = -(z^2+1)^n \mathcal{C}[M_n(z)^{-1} E(z) M_n(z)^{-1}].$$

Thus the pair alternates under each shift, returning after two shifts. This matters because norm preservation initially gives an intertwiner, not a commutant.

At $z = u - ik$, each local inverse has operator norm at most $[u^2 + (k-1)^2]^{-1/2}$. The norm multiplier per shift is at most

$$\left[\frac{u^2 + (k+1)^2}{u^2 + (k-1)^2}\right]^{n/2}.$$

These factors telescope. At the $n+1$ even-shift points, $\|E(u - 2ij)\| \leq B_n^n \delta$, $j = 0, \ldots, n$.

Interpolate the degree-n polynomial E. The sum of absolute leading-coefficient weights is $\sum_{j=0}^n [2^n j!(n-j)!]^{-1} = 1/n!$. The leading coefficient is $\widehat{Y} - \widehat{X}$; the interpolation weights at zero have numerators bounded by $(u+2n)^n$. Hence

$$\|Y - X\| \leq B_n^n \delta/n!, \qquad \|E(0)\| \leq B_n^n (u+2n)^n \delta/n!.$$

Since $M_n(0)$ is a unit phase times the cyclic permutation C of all $n+1$ sites, $\left\|[\widehat{X}, C]\right\| \leq K_n \delta$.

Let P_j be the normalized partial-trace conditional expectation making site j identity, where $j = 0$ is the source. These projections commute. Since $P_0 \widehat{X} = \widehat{X}$, cyclic rotation and telescoping

give $\left\|(I-P_j)\widehat{X}\right\| \le jK_n\delta$. Their joint projection is the scalar projection and vanishes on $\widehat{X}$. Therefore

$$1 \le \sum_{j=1}^{n} \left\|(I-P_j)\widehat{X}\right\|^2 \le \frac{n(n+1)(2n+1)}{6} K_n^2\delta^2.$$

This proves the bound. The two orientations $I - T_n^*T_n$ and $I - T_nT_n^*$ share the same singular defects. Stirling's formula gives the asymptotic exponent. □

3.2 Consequence for arbitrary growing memory

The witness has weight at most $m/2^n$ on the identity vector. Its energy and (5) imply

$$\frac{m}{2^n} \ge 1 - \frac{e_\epsilon}{g_n}, \qquad q_n = \log_2 m \ge n + \log_2(1 - e_\epsilon/g_n), \tag{7}$$

when $e_\epsilon < g_n$. This bound is explicit jointly in n, θ, ϵ, but can be weak at moderate accuracy.

At fixed θ, put $L_\epsilon = \log(1/\epsilon)$ and take

$$n = \left\lfloor \frac{(1-\eta)L_\epsilon}{4\log L_\epsilon} \right\rfloor, \qquad 0 < \eta < 1.$$

Then $g_n/\epsilon \to \infty$ and $q_n \ge n - o(1)$. Provided the stream reaches this cut,

$$\liminf_{\epsilon\to 0} \frac{q_{\mathrm{peak}} \log\log(1/\epsilon)}{\log(1/\epsilon)} \ge \frac{1}{4}. \tag{8}$$

Logarithms here are natural; q counts qubits. The branch-tail bounds give the same leading asymptotic for expected occupancy in the Bell-input experiment: use a threshold below g_n, where the low-energy mode count is one, then sum the memory tails.

At $N = \Theta_\theta(\log(1/\epsilon))$, including the quartic benchmark at fixed angle, this is $q_{\mathrm{peak}} = \Omega_\theta(N/\log N)$. It excludes $O(N^{1-\alpha})$ memory for every fixed $\alpha > 0$ in that regime, with arbitrary classical memory and processing allowed. It still permits an $N/\log N$-type receiver. The angle dependence is not uniform: the explicit g_n exposes it. No quartic memory lower bound as $\theta \to 0$ at fixed error follows from this argument.

3.3 General one- and two-qubit memory floors

Let γ_L be the minimum eigenvalue of H_L on the traceless subspace, including any further zero modes if present. The witness has weight at most m/D on the identity vector. Hence

$$e_\epsilon \ge \gamma_L(1 - m/D).$$

When $\gamma_L > 0$, this yields

$$\epsilon \ge \frac{3}{4}\left[1 - \sqrt{1 - \frac{4}{3}\gamma_L(1-m/D)}\right]. \tag{9}$$

At two carriers, the exact weak-angle gap is

$$\gamma_2 = 2\sin^4\theta - \sin^8\theta, \qquad \tan^4\theta \le \tfrac{1}{3}.$$

Thus every receiver with at most one persistent qubit after the second carrier has a lower-bound expansion $\theta^4/2 + O(\theta^6)$, independent of the number of subsequent carriers.

At three carriers, exact symbolic perturbation gives

$$\gamma_3 = \frac{7-\sqrt{33}}{2}\theta^4 + O(\theta^6). \tag{10}$$

Consequently every receiver with at most two persistent qubits after the third carrier has the error floor

$$\epsilon \;\geq\; \frac{7-\sqrt{33}}{8}\theta^4 - C\theta^6 \quad \text{for sufficiently small } \theta > 0 \tag{11}$$

for some finite constant C. The positive leading coefficient is approximately 0.15693. All classical histories and arbitrary subsequent operations are included.

An exact coefficient certificate is as follows. Set $t = \tan\theta$. The source-traceless leakage map has expansion $R(t) = tA + t^2B + O(t^3)$. Its first-order kernel is the five-dimensional SU(2)-scalar sector. Besides the identity, use

$$K_1 = \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2, \quad K_2 = \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_3, \quad K_3 = \boldsymbol{\sigma}_2\cdot\boldsymbol{\sigma}_3, \quad K_4 = \sum_{abc}\epsilon_{abc}\sigma_{1a}\sigma_{2b}\sigma_{3c}.$$

Their normalized Gram matrix is $G = \mathrm{diag}(3,3,3,6)$. Eliminating the first-order complement gives the effective metric

$$K^*B^*\big[I - A(A^*A)^+A^*\big]BK = \begin{pmatrix} 6 & 3 & -3 & 0 \\ 3 & 18 & 3 & 0 \\ -3 & 3 & 6 & 0 \\ 0 & 0 & 0 & 12 \end{pmatrix}.$$

Its eigenvalues relative to G are $(7-\sqrt{33})/2$, 2, 3, and $(7+\sqrt{33})/2$. The other 59 traceless leakage modes start at positive order θ^2. Analytic perturbation and angle-sign symmetry give the stated remainder. The accompanying exact-arithmetic script constructs the local Liouville maps and this matrix; it does not fit a numerical power law.

These floors exclude bounded one- and two-qubit solutions with arbitrarily small error at a fixed sufficiently weak angle. They do not exclude a receiver whose quantum memory grows sublinearly with N, or establish a large error at every practical tolerance.

The all-prefix bound proves (8), but is not sharp enough to settle every sublinear memory scaling. Spectra of a different local random eSWAP process cannot be substituted without a proved comparison. Integrable transfer-matrix eigenvalues also do not directly give these singular values: the homogeneous transfer problem can have Jordan blocks [5].

A conditional observation isolates the remaining loss in this route. A bound $\gamma_L \geq a(\theta)e^{-b(\theta)L}$ would allow a cut L proportional to N and exclude a uniform $o(N)$ memory claim as $\epsilon \to 0$ at fixed angle. Our proved bound has an additional $\log L$ in its exponent. No argument removing that factor is proved here. The padding identity in [A2] gives exponentially small *upper* bounds on certain losses; it cannot be used as a lower bound.

4 Exact erasure extension

Fix an input-independent retained set $A \subseteq \{1,\dots,N\}$. All other carriers are erased before receiver processing; the set is provided as a classical flag. Subsequent decoding on the retained outputs is unrestricted. Put $d = c^4$.

Theorem 3 (Retained-carrier law)**.** *Initialize $a_0 = 0$, $q_0 = 1$ and update chronologically by*

$$j \in A: \quad a_j = a_{j-1} + bq_{j-1}, \qquad q_j = (1-b)q_{j-1}, \tag{12}$$

$$j \notin A: \quad a_j = a_{j-1}, \qquad q_j = dq_{j-1}. \tag{13}$$

There is a physical decoder $\mathcal{D}_A$ for which

$$\mathcal{D}_A\mathcal{C}_A = a_N \,\mathrm{id} + (1-a_N)\mathcal{R}_{I/2}, \qquad d_\diamond(\mathcal{D}_A\mathcal{C}_A, \mathrm{id}) = \tfrac{3}{4}(1-a_N). \tag{14}$$

Every decoder of all retained outputs satisfies

$$\epsilon_{\mathrm{opt}}(A) \geq \tfrac{3}{4}(1-\sqrt{a_N}). \tag{15}$$

Proof. Propagate any initial source Pauli through the collision circuit and immediately trace an erased carrier. This equals tracing it at the end, since it never returns. Use normalized Hilbert–Schmidt inner products, and split the signal into its source-identity part P and source-traceless part Q. Let their squared norms be a and q.

Old P remains unchanged in the next collision. For any traceless source coefficient, partial SWAP sends its source-identity component to s^2 times the corresponding fresh-carrier Pauli. This export is orthogonal to old P because the fresh Pauli is trace zero. Its squared norm is $s^4 q$. Unitarity leaves $(1-s^4)q$ in the orthogonal Q sector. This proves (12), including arbitrary operator coefficients on older carriers. If the fresh carrier is erased, the source map is depolarizing with visibility c^2, so Q has squared norm $c^4 q$ and no new export survives. This proves (13).

For $m = |A|$, write the retained-output channel as

$$\mathcal{C}_A\big((I + \boldsymbol{u}\cdot\boldsymbol{\sigma})/2\big) = 2^{-m}\big(I + \textstyle\sum_k u_k B_k\big).$$

The preceding calculation and polarization give

$$2^{-m}\,\mathrm{Tr}(B_k B_l) = a_N \delta_{kl}, \qquad \mathrm{Tr}\, B_k = 0.$$

The average output is exactly maximally mixed. Its Petz map is therefore the dimension-normalized adjoint $\mathcal{D}_A = 2^{m-1}\mathcal{C}_A^*$. The Gram identity proves (14).

This adjoint is physical: start a mixed dummy, traverse the schedule backwards, apply $U^\dagger$ to each retained carrier, and at every erased position apply dummy depolarization of visibility c^2. A fresh mixed qubit, an inverse partial SWAP and discarding that qubit implement each replacement step.

Finally $\|\mathcal{C}_A(\sigma_k)\|_1 \leq 2\sqrt{a_N}$ by Hilbert–Schmidt Cauchy–Schwarz. A decoder contracts trace norm. Its entanglement fidelity with the identity is consequently at most $(1+3\sqrt{a_N})/4$. Testing its normalized Choi state with the Bell projector gives (15). The upper/lower error ratio is $1+\sqrt{a_N} \leq 2$ whenever the lower bound is nonzero. □

4.1 Independent flagged loss

Suppose each carrier is retained independently with probability $p = 1-\ell$. Independence from the payload and the other erasures is essential. Define

$$f = p(1-b) + (1-p)d, \qquad \bar{a}_N = pb\frac{1-f^N}{1-f}.$$

Averaging the exact recurrence gives this expression. The flagged branchwise Petz decoder has visibility $\bar{a}_N$ and error $\frac{3}{4}(1-\bar{a}_N)$. Any decoder of the complete flagged output has error at least

$$\tfrac{3}{4}(1-\sqrt{\bar{a}_N}),$$

because $\mathbb{E}\sqrt{a_N} \leq \sqrt{\mathbb{E}a_N}$. These are unconditional channel guarantees, including the loss flags.

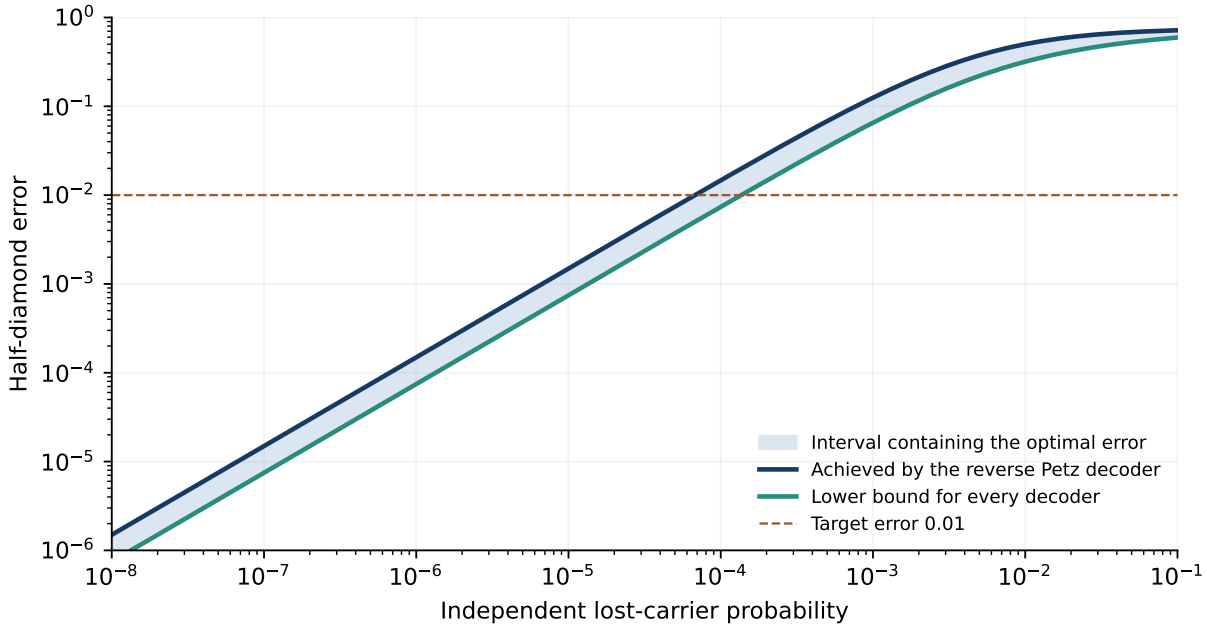


Figure 1: Exact analytic infinite-count bounds at $\theta = 0.1$. The shaded interval contains the optimal half-diamond error for independent flagged erasure. Neither boundary is asserted to be the exact optimum. The horizontal line is an error target of 0.01. This plot evaluates formulas; it is not a simulation of a large quantum register.

As $N \to \infty$,

$$\bar{a}_\infty = \frac{(1-\ell)\sin^4\theta}{(1-\ell)\sin^4\theta + \ell(1-\cos^4\theta)}. \tag{16}$$

Acquisition contributes at fourth order, while losing a carrier attenuates the still-encoded signal at second order. In particular,

$$\ell = \kappa\theta^2 \quad \Longrightarrow \quad \bar{a}_\infty \longrightarrow \frac{1}{1+2\kappa} \qquad (\theta \to 0). \tag{17}$$

If $\ell/\theta^2 \to \infty$, the optimal error tends to $3/4$, even with arbitrarily many encounters and unrestricted final decoding.

For $0 < \epsilon < 3/4$, put $A_\epsilon = (1 - 4\epsilon/3)^2$, $t = 4\epsilon/3$, and $g = 1 - c^4$. A necessary asymptotic condition for error at most ϵ is

$$\ell \le \frac{(1-A_\epsilon)b}{A_\epsilon g + (1-A_\epsilon)b}. \tag{18}$$

A sufficient condition leaving a strictly positive finite-count margin for the Petz decoder is

$$\ell < \frac{tb}{(1-t)g + tb}. \tag{19}$$

Both scale as $\ell = \Theta(\epsilon\theta^2)$ as angle and error become small. A finite count is then obtained from

$$\tfrac{3}{4}[1 - \bar{a}_\infty + \bar{a}_\infty f^N] \le \epsilon.$$

Loss probability	Petz error $N = 43\,462$	Any-decoder bound $N = 43\,462$	Any-decoder bound $N \to \infty$
0	0.00999962	0.00501659	0
10^{-5}	0.0113890	0.00571628	0.000747643
10^{-4}	0.0236813	0.0119356	0.00737798
10^{-3}	0.128466	0.0672476	0.0653091
10^{-2}	0.501397	0.318199	0.318198

Thus losing one carrier in a thousand already prevents error 0.01 at $\theta = 0.1$, regardless of how long this independently erased stream is collected.

4.2 Keeping only an untouched subset

For a budget of at most m untouched retained carriers, a_N is maximized by retaining the first m. Exchange an adjacent erased–retained pair for retained–erased. The acquired coefficient increases by $b(1-d)q$, while the later residual coefficient is unchanged. Repeated exchanges give

$$a_N \le 1 - (1-b)^m.$$

Consequently every decoder of any such subset needs

$$m \ge \frac{\log[1/\{1-(1-4\epsilon/3)^2\}]}{-\log(1-b)} \tag{20}$$

to achieve error $\epsilon < 3/4$. Retaining the first m and applying the inverse reader suffices when

$$m \ge \frac{\log(3/(4\epsilon))}{-\log(1-b)}. \tag{21}$$

The retained-carrier requirement in this class is therefore $\Theta(\theta^{-4}\log(1/\epsilon))$. Input-independent random selection does not improve the lower bound. The exchange argument optimizes the exact Gram/Petz coefficient; no claim is made that it exactly orders every subset by its optimal decoding error. In particular, this theorem does not cover coherent processing before discard.

5 Constructions tested and their actual scopes

Three constructors began separately with the channel formulation, an operational input/output specification, and an operator/MPO formulation. Separate prompts did not remove shared model training or make the attempts statistically independent. Mechanisms were exchanged only after concrete derivations were available.

The operational route recovered the inverse reader independently. It also ruled out a particular pure Bell dark-state absorber. A fresh tracial carrier can be purified by an inaccessible qubit; one source collision creates source–purifier correlation. Receiver operations cannot change that marginal, so they cannot restore an exact pure Bell state between source and receiver. This is a local obstruction, not a general memory theorem.

The folded inverse tensor is

$$V = cI - 2is\Phi, \qquad V^\dagger V = c^2 I + 4s^2\Phi,$$

with singular values c and $\sqrt{1+3s^2}$. It is not an isometry. The earlier Astra note [A2] already proves that a particular two-qubit folded statistic is not causal and quantifies the postselection cost. A new constructive attempt completed this tensor to the valid instrument

$$K_k = \tfrac{1}{2}V(P_k \otimes I), \qquad k = 0,1,2,3, \qquad \sum_k K_k^\dagger K_k = I,$$

where P_k are the four Paulis. The outcome at the first step can be corrected using the initial receiver Bell pair, but discarding the processed carrier already loses information. With $x = \sin^2\theta$, the distinguishability of two opposite source inputs in the full source-plus-receiver state becomes

$$D_1 = \frac{x + \sqrt{4(1-x)^4 + 4x - 3x^2}}{2} < 1 \qquad (0 < x < 1).$$

All corrected first branches agree, and the record is input-independent. No subsequent operation, even with the source granted, can increase that distinguishability. Thus this particular first-step discard forces final error at least $(1 - D_1)/2 = \theta^2/2 - 11\theta^4/12 + O(\theta^6)$. Independent two-step checks also refute the proposed Pauli-frame closure. These statements exclude the specified mechanism, not every small receiver.

Other boundary calculations encountered here, including measured Schur compression, pair-sector pinching and purified-dummy block readers, were already treated in [A2]. They are not new results from this run. Pure-state tensor-network compression [4] also cannot be applied merely from a small mixed-state MPO bond; its operational premises must be met.

6 Verification, provenance and limitations

The crossing/intertwiner identities, interpolation, variance normalization and finite spectral bound were checked independently at prefixes $n = 1, \ldots, 4$ and five positive or negative values of u. All 20 cases passed; the largest relative identity/interpolation error was 3.4×10^{-10}, arising from degree-four interpolation cancellation. The crossing identities themselves had tighter residuals. These checks corroborate the analytic proof and do not replace it. The exact three-carrier coefficient matrix was also recomputed in rational symbolic arithmetic.

The retained-carrier recurrence was checked against dense channel matrices for every subset at $N = 1, \ldots, 5$ and $\theta \in \{0.1, 0.4, 0.9\}$: 186 cases, with maximum Gram-entry discrepancy 4.44×10^{-16}. The proof above is independent of those floating-point checks. Large-count numerical rows evaluate the analytic recurrence only.

Additional finite singular-value calculations cover the full bath operator space through four carriers and the SU(2)-scalar operator sector through seven. The latter is a restricted invariant sector, not automatically the entire smallest-gap spectrum. Closure residuals are at floating-point scale. These computations do not establish large-N behavior. The archive retains code and numerical receipts with that scope.

The quantitative crossing bound, sharpened witness coefficient, finite-memory floors and erasure extension were derived and combined during this investigation, using established recovery and R-matrix machinery. An exhaustive priority claim is not made. The erasure result concerns loss before receiver access and is not substituted for a general compression obstruction. The all-output operation, earlier general profile framework and prior restricted failures retain their attribution.

Resolution status. No causal sublinear-memory decoder with the requested guarantee has been constructed. The general $\Omega_\theta(N/\log N)$ lower bound does not match the known N-qubit reader or settle the joint weak-angle limit. The fixed endpoint remains unresolved. What remains is either a physical causal compression with the full recovery and resource guarantees, or a sufficiently sharp general converse closing this gap. The present lower bound is not relabeled as a full resolution.

Sources

[**A1**] Astra research knowledge, *Tracial acquisition: partial-SWAP decoder*, return dated 5 November 2026. Repository `harjassand/astra-research-knowledge`. Pinned commit: `fef3640a8b1dc5475674a006de417f652f3baf3c`. Governing source: `research/returns/2026-11-05-p2-tracial-acquisition/package/partial_swap_theory.txt`. Retrieved as research material, not represented as a peer-reviewed publication.

[**A2**] *Causal memory for the homogeneous tracial partial-SWAP stream: Proved bounds, physically specified candidates, and the unresolved tradeoff*, 5 November 2026, 13-page Astra research note. Existing file `weak_stream_memory.pdf`. Identifier: `libfile_66a8b33dfd988191b76f181429691b90`. The prefix framework and restricted obstructions cited here are prior material.

[**1**] M. Ziman et al., *Quantum homogenization*, Phys. Rev. A 65, 042105 (2002). https://arxiv.org/abs/quant-ph/0110164.

[**2**] J. M. Renes, *Better bounds on optimal measurement and entanglement recovery, with applications to uncertainty and monogamy relations*, Phys. Rev. A 96, 042328 (2017). https://arxiv.org/abs/1707.01114.

[**3**] C. C. Aw, L. H. Zaw, M. Balanzó-Juandó and V. Scarani, *Role of dilations in reversing physical processes: Tabletop reversibility and generalized thermal operations*, PRX Quantum 5, 010332 (2024). https://arxiv.org/abs/2308.13909.

[**4**] G. Bai, Y. Yang and G. Chiribella, *Quantum compression of tensor network states*, New J. Phys. 22, 043015 (2020). https://doi.org/10.1088/1367-2630/ab7a34.

[**5**] P. W. Claeys, J. Herzog-Arbeitman and A. Lamacraft, *Correlations and commuting transfer matrices in integrable unitary circuits*, SciPost Phys. 12, 007 (2022). https://arxiv.org/abs/2106.00640.